FLUOR DANIEL

PIPE SUPPORTS

PURPOSE

This practice establishes guidelines and recommended procedures for the design of pipe supports.

SCOPE

This practice includes the following major sections:

- GENERAL
- DESIGN LOADS
- DESIGN OF PIPE SUPPORT COMPONENTS
- REFERENCES
- ATTACHMENTS

APPLICATION

This practice applies to all structures as described herein as pipe supports.

GENERAL

The term "pipe supports" describes a class of structures ranging from small supports carrying light utility lines to the main multilevel pipeways loaded with air coolers. There are 6 basic types of supports, as follows:

- Strutted concrete main pipeways
- Unstrutted concrete secondary pipeways
- Strutted steel main pipeways
- Unstrutted steel secondary pipeways
- Miscellaneous single column concrete or steel supports as required throughout the plant
- Sleeper supports

Usually, pipe supports, strutted or unstrutted, are designed as rigid frames, bents, in the transverse direction. In the longitudinal direction, strutted pipe supports may be designed with the longitudinal struts either acting with the columns transmitting all longitudinal loads to vertical bracing, or as a continuous rigid frame. The design approach used will depend on the job criteria. Unstrutted pipe supports are usually designed as cantilever members in the longitudinal direction. Longitudinal and transverse directions are defined in the attached figures.

The spacing of supports is based on the allowable span for piping and electrical cable tray being supported. Typical support spacings are 20 to 25 feet; however, the bent spacing can vary from 10 to 40 feet with intermediate beams being supplied when the support spacing exceeds 20 to 25 feet. Pipe bridges are typically used when the spacing between support bents exceeds 40 feet, which normally occurs at road crossings. Clearances over, under, and around pipe supports are an important consideration in their design. Normally, these clearances are established in the project design criteria. Due consideration should be given to clearance requirements and existing and proposed interferences prior to performing pipe support design calculations.

The construction material is generally established by site conditions, fireproofing requirements, procurement limitations, and client preferences. Usually, if fireproofing is not required, steel is the most economical and easily erected and modified material. However, for large pipe supports that require fireproofing, precast concrete may be the most economical alternative. Shop fireproofing of steel members with connection areas left open for field fireproofing or connections outside of fireproofing, also can be an economical alternative. The requirements for fireproofing are defined in the fireproofing specifications of the project.

Computer programs and spreadsheets are available for the analysis and the design of pipe supports; their use is greatly encouraged, but should be coordinated through the project Lead Structural Engineer.

DESIGN LOADS

The design loads discussed below include gravity loads and lateral loads. Also, combinations of these loads are defined.

Gravity Loads

Gravity loads include piping, electrical, structural, and equipment loads.

Piping

An average pipe deck load of 40 psf (Pounds Per Square Foot) should be used for major pipe supports. This corresponds to an equivalent load of 8 inch pipes full of water, spaced at 15 inches c/c, which is considered to be an average operating load condition for pipe supports. The empty load condition may be taken as 60 percent of the operating load condition. Preliminary piping and process information should be consulted to determine if a heavier or possibly a lighter load should be considered. A concentrated load should be added at pipes which are at least 2 sizes larger than the average on the support. This concentrated load can be calculated using the tables attached to this practice with the following formula:

$$P = s(w - pd)$$

where

P	=	Concentrated load
s	=	Support spacing
w	=	Weight of pipe per unit length
p	=	Pipe deck load
d	=	Pipe diameter.

When analyzing existing pipe supports, the actual piping supported on the pipe supports should be considered. It is not necessary that the piping be applied to the structure as individual concentrated loads except as described above. A uniformly distributed load representative of the existing piping is preferable. A minimum operating load of 25 psf should be used for piping on any pipe support, new or existing. Also, empty and future areas on pipe supports should be considered loaded as described above.

For large vapor and flare lines, it should be established whether or not the line will be hydrotested in place. Also, the normal operating weight of the line should be established, since it is usually 10 percent to 30 percent of the full of water weight of the pipe. This is especially important when adding to existing pipe supports.

Electrical

The electrical group should be consulted to determine the approximate weight and location of electrical trays or conduits. A minimum weight of 20 psf should be used for single level trays, and 40 psf for double level trays.

Structural

The weight of all structural members, including fireproofing, should be considered in the design of the support. Usually, the calculation of fireproofing weight is made by adding 2 inches of concrete cover to the nominal column, or beam flange, width and depth, with the exception that the top flanges of beams are usually left exposed.

Equipment

Weights for equipment such as air coolers, including weights of all associated platforms, ladders, walkways, and headers, should be obtained from suppliers engineering data and piping layouts. For estimating purposes, typical air cooler loads are given in the Attachment 03.

Unusual Loads

Special consideration should be given to unusual loads such as large valves, unusual piping, or electrical configurations.

Occasionally, access platforms are supported on pipe supports. Platforms should be designed for live loads specified in the project requirements.

Lateral Loads

The lateral loads discussed below include wind and earthquake, friction and anchor loads. These loads are described as acting in the transverse direction or the longitudinal direction. Refer to the attached figures for definitions of transverse and longitudinal directions.

Wind And Earthquake Loads

Transverse wind loads will be applied to pipe supports as described in Structural Engineering Practice 670.215.1215: Wind Load Calculation.

Longitudinal wind loads are usually small compared to other longitudinal loads and can be disregarded unless air coolers or other unusual conditions are present.

Longitudinal and transverse earthquake loads will be applied according to Structural Engineering Practice 670.215.1216: Earthquake Engineering.

Friction Loads

Friction loads caused by hot lines sliding across the pipe support during startup and shutdown are assumed to be partially resisted by adjacent cold lines. Therefore, in order to provide for a nominal unbalance of friction loads acting on a pipe support, a resultant longitudinal friction load equal to 10 percent of the total pipe weight tributary to that pipe support is assumed for main pipe supports. At individual supports (transverse beams), a horizontal longitudinal load will be considered to act as a uniformly distributed load across the member as follows:

- 10 percent of the total pipe weight for number of pipes > 7
- 20 percent of the total pipe weight for number of pipes = 4, 5, or 6
- 30 percent of the total pipe weight for number of pipes = 1, 2, or 3

For a given support, if considering only larger lines and ignoring smaller lines, resulting in greater loads, these forces and associated friction coefficients shall be used instead of considering all the lines.

Anchor Loads

Piping anchors (and guides) cause expansion movement to occur at desired locations in a piping system. The Pipe Stress Engineer is responsible for locating anchors and providing anchor loads. For this reason, it is important that the Structural Engineer communicate with the Pipe Stress Engineer prior to starting and during the design of any pipe support.

Anchor loads are usually small and adjacent pipes will transfer the load laterally to the longitudinal beam struts. It is normally preferred to either have the anchors staggered along the pipeway so that each support has only 1 or 2 anchors, or to anchor all pipes on 1 braced support. Special consideration should be given to pipe supports on which all or most of the lines are anchored, or on which significant anchor loads are anticipated.

Anchor loads have 2 components, thermal and friction. The friction component is related to the friction loads defined above. Engineering judgment will be exercised in determining to what extent the friction and anchor loads are to be combined to design a pipe support.

Since anchor loads are normally not available until the latter stages of a project, steel pipe supports may be designed without considering anchor loads. When the anchor loads become available, supports will be checked individually for the actual anchor load and reinforced if necessary.

Since modifications to concrete pipe supports after construction are costly and time consuming, an imaginary anchor load will be considered in the design when actual anchor loads are not available. This imaginary anchor load will be equal to 2 kips for beam spans greater than 15 feet, and 1 kip for beam spans less than 15 feet. This load will be applied at the 1/4 span locations of the beam at each level. Thirty percent of the imaginary load is to be considered the thermal component, and 70 percent is the friction component.

Load Combinations

Pipe supports will normally be designed to resist the following combinations of loads:

- Gravity loads (empty, operating, and test).
- Gravity loads (empty) + transverse wind loads or earthquake loads.
- Gravity loads (operating) + transverse wind loads or earthquake loads + thermal component of anchor loads.
- Gravity loads (operating) + friction loads + thermal component of anchor loads.
- Gravity loads (operating) + anchor loads (friction and thermal components).

It is assumed that friction loads, including the friction component of anchor loads, do not occur at the same time as wind or earthquake loads. Engineering judgment or project design specifications may dictate otherwise. Also, project requirements sometimes stipulate that test loads be combined with a reduced wind load.

Design Stresses

Usually, allowable steel stresses may be increased 1/3 for load combinations that include wind or earthquake loads; however, the allowable stress increases must be specified in the design specifications of the project.

When applying ultimate strength load factors for concrete design, all gravity loads will be considered as dead loads. Although platform loads are mostly live loads, it is acceptable to include them with gravity loads as long as they are small; less than 10 percent of the total gravity load on the member. Platform framing members should be designed for live loads with live load factors. Friction and anchor loads should be considered as dead loads for ultimate strength design.

Deflections Of Pipe Supports

The deflection of structural members in a pipe support is an important consideration in the design of the piping system. Whether the deflection is that of a transverse beam due to piping anchors or lateral deflection of the bent due to wind loads, the criteria for the deflection will be as specified in project requirements. However, where the Pipe Stress Engineer indicates that the deflection of a particular anchor is critical, where multiple anchors are located on the same support, or where there is an absence of other lines to provide restraint, the deflection should be calculated and reviewed with the Pipe Stress Engineer for concurrence.

DESIGN OF PIPE SUPPORT COMPONENTS

The components of pipe supports discussed below include rigid frames (bents), longitudinal struts, vertical bracing, connections, and foundations. Refer to the attached figures for reference. Also discussed are the structural elements of pipe bridges and small supports.

The design of pipe support components is primarily based on stress constraints. At times, deflections and settlement of pipe supports merit special consideration, thereby affecting the design of the pipe support components. In such cases, the design of the pipe support will be coordinated with the Pipe Stress Engineer as mentioned above to ensure that movement constraints are met.

Rigid Frames (Bents)

Normally, a stiffness analysis of a transverse bent composed of transverse beams and columns is performed to determine all forces, reactions, and displacements produced by the loads and load combinations given above.

For analysis of concrete frames, approximate slenderness effects such as moment magnification or a second order analysis may be performed. Appropriate stiffness values for the beams and columns should be used according to ACI (American Concrete Institute) 318 in either type of analysis.

Precast concrete bents will be analyzed for handling stresses induced from being transported and lifted.

Transverse Beams

The beam must be designed to resist all forces, moments, and shears calculated from the above analysis.

For the flexural design of steel beams, the unbraced length of the compression flange should be considered 1/3 of the total span. However, for axial loads, the total span of the beam should be used for the effective length and modified by the appropriate effective length factor for each direction. This factor should be equal to 1.0 for the weak direction of the beam. In the strong axis for moment connected ends, the effective length factor should be 0.65.

Under normal loading conditions, torsional effects need not be considered since the pipe supported by the beam limits deflection and rotation of the beam to the extent that torsional stresses are minimal. However, torsion should be considered on an individual basis when unusually large loads such as large anchor loads are applied to the beam flange.

Intermediate transverse beams are sometimes required to reduce the span for smaller pipe and cable trays. Also, they are required at pipe bridges. Generally, intermediate transverse beams are supported by struts or the chords of pipe bridge trusses. They are designed as simply supported beams.

Columns

The columns must be capable of resisting all forces, moments and shears calculated from the rigid frame analysis. The frame analysis should be made using the following column base conditions:

- Steel pipe supports

Strutted - fixed base in both the transverse and longitudinal directions, or pinned base in both the transverse and longitudinal directions, with the major axis of the column in the transverse direction. In general, the fixed base condition results in a smaller superstructure and a larger foundation with smaller lateral deflections. The pinned base condition results in a larger superstructure and smaller foundations with larger lateral deflections.

Unstrutted - fixed bases in both directions, with the major axis of the column in the longitudinal direction. A common design concept is to provide bracing in the transverse direction.

- Concrete pipe supports

Fixed at the top of the socket (for socket type footings) or the base plate.

The effective length factors for the design of columns will be as follows:

- Longitudinal

Strutted - Table C-C2.1, Pages 5 - 135, AISC (American Institute for Steel Construction) ASD 9th Ed.

Unstrutted - K = 2.0 or lesser value approved by the Project Lead Structural Engineer.

- Transverse

Steel - Table C-C2.1, Pages 5 - 135, AISC ASD 9th Edition, or Figure C-C2.2, pages 5 - 137, AISC ASD 9th Edition, or another method approved by the Project Lead Structural Engineer.

Concrete - Figure R10.12.1, ACI-318-95.

Columns for concrete pipe supports should be 18 inches square minimum.

A design check should be performed for the temporary lifting of precast concrete bents.

Longitudinal Struts

In areas where gravity loading on struts is anticipated, beam struts should be used.

Beam struts should be designed for the greater of 50 percent of the gravity loading on the most heavily loaded transverse beam or the actual loading. The 50 percent loading accounts for the usual electrical conduits and piping takeoffs. This loading should not be added to the design load for the column or footing, since pipes contributing to the load on the struts reduces the load on the transverse beams. Prior to issuing any pipe support drawings as AFC (Approved for Construction), the Design Engineer should check piping drawings to verify that any struts subjected to unusually large loads have been given special consideration.

Longitudinal struts will be designed to resist axial forces produced by longitudinal loads. For normal conditions, longitudinal loads may be assumed to be transmitted to the struts at each column without reconsidering column bending in combination with the rigid frame analysis. However, if the vertical dimension between transverse beam and the strut in question is large (exceeding 3 feet), or large anchor loads occur on the transverse beam, the column stresses must be reconsidered.

Vertical Bracing

Vertical bracing may be used to transmit longitudinal loads from the struts to the foundations. K-bracing (inverted chevron bracing) is most often used for this purpose. Normally, the maximum spacing of braced bays should be limited to 150 feet. Operating access is an important consideration when locating bracing. The Structural Engineer will coordinate the placement of bracing with the Piping and Electrical groups. Slotted strut connections are sometimes used to isolate the longitudinal loads on a run of pipe support to specific braced bays. The locations of slotted connections should be reviewed with the Pipe Stress Engineer.

Connections

Connection details described below include moment connections, base plates, and other connections commonly used in pipe support design.

Moment Connections

Moment connections shall be designed in accordance with Structural Engineering Practice 670.215.1209: Bolted End Plate Moment Connections.

Base Plates

Base plates will be designed in accordance with Structural Engineering Practice 670.215.1208: Base Plate Design Criteria, with anchor bolts designed in accordance with Structural Engineering Practice 670.215.1207: Anchor Bolt Design Criteria.

Other Connections

Bracing and framed beam connections will be designed in accordance with the AISC Manual. These connections will be as specified and detailed in the project standard drawings; however, where the standard details are not appropriate or adequate, proper details will be shown on the construction drawings.

Special attention will be given to standard shear connections used in situations with high tension loads such as struts of large pipe supports, and especially when dealing with longitudinal air cooler loads. The connection angles should be checked according to the Hanger Type Connections section of the AISC Manual.

Suggested references for unusual steel connections are Salmon and Johnson, Steel Structures Design and Behavior and Blodgett, Design of Welded Structures.

For connections between steel struts and concrete columns, which are usually required when using precast concrete bents, some type of insert will be required. Embedded plates cast into the concrete member with welded rebar or headed studs for anchorage or through bolts with sleeves cast in bents may be used. Expansion anchors are not preferred. In addition, the PCI Design Handbook describes the design of various types of connections. The selected connection detail should be used uniformly throughout the project in order to be economical.

Foundations

The type of foundations to be used will be dictated by the site conditions. Foundations will be designed using the support reactions at the column bases from the rigid frame analysis and the braced bay. Foundation design parameters are normally stated in the project design specifications.

The stability ratio shall be checked for the most critical overturning condition. For high wind areas, the empty load condition generally controls. In high seismic zones, the heaviest load results in higher overturning forces. When a rigid frame is supported on 2 or more foundations, the stability of the entire system will be considered. Engineering judgment will be used to determine if the stability of the foundation system or an individual foundation within the system is more critical.

Pipe Bridges

Prior to making a pipe bridge design, the Design Engineer should verify with the Piping group where pipes will be supported on the bridge. A pipe bridge should be designed as individual components including vertical trusses (or girders), horizontal trusses, and bridge bents. Refer to the Attachments 04 and 5. A computer space frame solution for a pipe bridge is generally not required or recommended due to the excessive amount of time required to make the computer model. However, where complex loading or unusual geometric configurations are present, a space frame solution is desirable.

Vertical Trusses

The vertical truss of a pipe bridge should be designed as a plane truss supporting gravity loads only. In many cases, the vertical truss can be fabricated as a single shop welded unit. The connections of the truss should be designed to accommodate field assembly of the truss as individual members or as a unit.

Horizontal Trusses

Horizontal trusses should be designed as plane trusses to resist all lateral loads applied to the truss such as wind or lateral earthquake loads. Also, consideration should be given to providing lateral support to intermediate transverse beams, especially where anchors or large diameter pipes are present.

Bridge Bent

The bridge bent is designed similar to a typical pipe support bent with the exception that truss loads are applied as concentrated loads to the bent. Should the member sizes of the bridge bent become excessive, transverse vertical bracing should be used with the approval of the client.

Small Supports

Small supports include T-supports, sleeper supports, and miscellaneous pipe supports requested by the Piping/Pipe Stress groups. Refer to Attachments 04 and 05. These supports usually require a minimal amount of structural analysis; however, they often require a significant amount of design time to ensure that geometric constraints are satisfied.

T-Supports

T-supports are usually single columns with short cantilevered beams attached to support piping or electrical conduit/cable trays. The effective length factor, K, of the column in both the transverse and longitudinal direction, is generally equal to 2.0. Where engineering judgment is exercised to allow a lower value for K, especially in the longitudinal direction (in the weak axis of the column), the value and base assumptions will be approved by the Project Lead Structural Engineer. Guide to Pipe Support Design by C. V. Char provides more details on effective length factor.

Sleeper Supports

Sleeper supports are used to elevate pipes at low levels above the ground. Their design is relatively simple; however, close coordination with pipe stress is required to ensure that anchor loads are properly handled and settlement sensitive areas addressed.

Miscellaneous Pipe Supports

Most miscellaneous pipe supports such as base ells and hangers are provided by the Piping group; however, there are cases where the Structural group is required to provide these supports, especially in the case of hold-downs at compressors. When designing small individual pipe supports, the usual safety factors applied to larger structures do not adequately reflect the uncertainty of the loading that the small support will subjected to. Engineering judgment should be exercised to ensure a safe and economical design.

REFERENCES

ACI (American Concrete Institute) 318-95

AISC (American Institute for Steel Construction) ASD 9th Edition.

PCI (Prestressed Concrete Institute). Precast and Prestressed Concrete. PCI Design Handbook. Third Edition, Chicago, 1985.

Blodgett, Omer W. Design of Welded Structures. Eighth Printing, The James F. Lincoln Arc Welding Foundation. Cleveland, Ohio, 1976.

Char, C. V. Hydrocarbon Processing. Guide to Pipe Support Design. Vol. 58, 1979.

Salmon, Charles G. and John E. Johnson. Steel Structures Design and Behavior, 2nd Edition, Harper & Row, Publishers, New York 1980.

Structural Engineering
Practice 670.215.1207: Anchor Bolt Design Criteria

Structural Engineering
Practice 670.215.1208: Base Plate Design Criteria

Structural Engineering
Practice 670.215.1209: Bolted End Plate Moment Connections

Structural Engineering
Practice 670.215.1215: Wind Load Calculation

Structural Engineering
Practice 670.215.1216: Earthquake Engineering

Structural Engineering
Practice 670.215.1231: Drilled Pier Foundations

ATTACHMENTS

Attachment 01:
Weights Of Pipe

Attachment 02:
Weight Of Piping Insulation

Attachment 03:
Typical Air Cooler Loads

Attachment 04:
Typical Piperack Configuration

Attachment 05:
Figure 1. Typical Pipe Bridge
Figure 2. Miscellaneous Pipe Supports

Attachment 06:
Sample Design 1: Steel Piperack Design

Attachment 07:
Sample Design 2: Concrete Piperack Design

Attachment 08:
Sample Design 3: Concrete Piperack Design With Seismic Design

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WEIGHTS OF PIPE

		-- ST (Standard Weight) --				-- XS (Heavy Weight) --				--XX 160 (Xtra Heavy Weight)--			
D (in.)	OD (in.)	t (in.)	We (plf)	Ww (plf)	Wf (plf)	t (in.)	We (plf)	Ww (plf)	Wf (plf)	t (in.)	We (plf)	Ww (plf)	Wf (plf)
1	1.320	0.133	1.7	0.4	2.1	0.179	2.2	0.3	2.5	0.358	3.7	0.1	3.8
1-1/2	1.875	0.145	2.7	0.9	3.5	0.200	3.6	0.7	4.3	0.375	6.0	0.4	6.4
2	2.375	0.154	3.7	1.5	5.1	0.218	5.0	1.3	6.3	0.400	8.4	0.8	9.3
3	3.500	0.215	7.6	3.2	10.8	0.300	10.3	2.9	13.1	0.600	18.6	1.8	20.4
4	4.500	0.237	10.8	5.5	16.3	0.337	15.0	5.0	20.0	0.674	27.6	3.4	30.9
5	5.600	0.258	14.7	8.8	23.5	0.375	20.9	8.0	29.0	0.750	38.9	5.7	44.6
6	6.625	0.280	19.0	12.5	31.5	0.432	28.6	11.3	39.9	0.864	53.2	8.2	61.4
8	8.625	0.322	28.6	21.7	50.3	0.500	43.4	19.8	63.2	0.906	74.8	15.8	90.6
10	10.750	0.365	40.5	34.2	74.7	0.500	54.8	32.4	87.1	1.125	115.8	24.6	140.3
12	12.750	0.375	49.6	49.0	98.6	0.500	65.5	47.0	112.5	1.312	160.4	34.9	195.3
14	14.000	0.375	54.6	59.8	114.4	0.500	72.2	57.5	129.7	1.406	189.3	42.6	231.9
16	16.000	0.375	62.6	79.2	141.8	0.500	82.8	76.6	159.4	1.593	245.3	55.9	301.2
18	18.000	0.375	70.7	101.3	171.9	0.500	93.5	98.4	191.9	1.718	299.0	72.2	371.2
20	20.000	0.375	78.7	126.1	204.8	0.500	104.2	112.9	227.1	1.968	379.4	87.8	467.2
22	22.000	0.375	86.7	153.7	240.4	0.500	114.9	150.1	265.0	*1.000	224.5	136.1	360.6
24	24.000	0.375	94.7	184.0	278.7	0.500	125.6	180.0	305.6	2.343	542.4	127.0	669.4
26	26.000	0.375	102.7	217.0	319.7	0.500	136.3	212.7	349.0	*1.000	267.3	196.0	463.3
28	28.000	0.375	110.7	252.7	363.5	0.500	147.0	248.1	395.1	*1.000	288.6	230.1	518.7
30	30.000	0.375	118.8	291.2	409.9	0.500	157.7	286.2	443.9	*1.000	310.0	266.8	576.8
32	32.000	0.375	126.8	332.4	459.1	0.500	168.4	327.1	495.4	*1.000	331.4	306.3	637.7
34	34.000	0.375	134.8	376.3	511.1	0.500	179.1	370.6	549.7	*1.000	352.8	348.5	701.3
36	36.000	0.375	142.8	422.9	565.7	0.500	189.8	416.9	606.7	*1.000	374.2	393.4	767.6
42	42.000	0.375	166.9	579.1	746.0	0.500	221.8	572.1	793.9	*1.000	438.3	544.5	982.8

D = Nominal Diameter
OD = Outside Diameter
t = Wall Thickness
We = Empty Weight of Pipe
Ww = Weight of Water
WF = Weight of Pipe Full of Water
* = Maximum Stock Size

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WEIGHT OF PIPING INSULATION

To determine the weight per foot of any piping insulation, use the pipe size and nominal insulation thickness to find the insulation weight factor, *F*, in the chart shown below. Then, multiply *F* by the density of the insulation in pounds per cubic foot.

Example: **For 4 inch pipe with 4 inch nominal thickness insulation, *F* = 0.77. If the insulation density is 12 pounds per cubic foot, then the insulation weight is 0.77 x 12 = 9.24 lb/ft.**

Note!!! **Usual insulation density is 12 pcf.**

Nominal Pipe Size	Nominal Insulation Thickness										
	1"	1-1/2"	2"	2-1/2 "	3"	3-1/2"	4"	4-1/2"	5"	5\1/2"	6"
1	0.057	0.10	0.16	0.23	0.31	0.40	—	—	—	—	—
1-1/2	0.066	0.11	0.21	0.29	0.38	0.48	—	—	—	—	—
2	0.08	0.14	0.21	0.29	0.37	0.47	0.59	—	—	—	—
3	0.10	0.17	0.25	0.34	0.44	0.56	0.68	0.81	—	—	—
4	0.13	0.21	0.30	0.39	0.51	0.63	0.77	0.95	1.10	—	—
5	0.15	0.24	0.34	0.45	0.58	0.71	0.88	1.04	1.20	—	—
6	0.17	0.27	0.38	0.51	0.64	0.83	0.97	1.13	1.34	—	—
8	—	0.34	0.47	0.66	0.80	0.97	1.17	1.36	1.56	1.75	—
10	—	0.43	0.59	0.75	0.93	1.12	1.32	1.54	1.76	1.99	—
12	—	0.50	0.68	0.88	1.07	1.28	1.52	1.74	1.99	2.24	2.50
14	—	0.51	0.70	0.90	1.11	1.34	1.57	1.81	2.07	2.34	2.62
16	—	0.57	0.78	1.01	1.24	1.49	1.74	2.01	2.29	2.58	2.88
18	—	0.64	0.87	1.12	1.37	1.64	1.92	2.21	2.51	2.82	3.14
20	—	0.70	0.96	1.23	1.50	1.79	2.09	2.40	2.73	3.06	3.40
24	—	0.83	1.13	1.44	1.77	2.10	2.44	2.80	3.16	3.54	3.92

TYPICAL AIR COOLER LOADS

(For estimating only)
(Loads shown on pipe support column)

Span Length, L	20'	25'	30'
Dead Load	35 k/col	42 k/co	50 k/col
Live Load	3.5 k/col	4 k/col	5 k/col
Wind Load:			
Transverse Shear	5 k/col	5.5 k/col	6 k/col
Wind Couple, Vertical	+/- 4.5 k/col	+/- 4.5 k/col	+/- 4.5 k/col
Longitudinal Shear (at braced bay only)	18 k/bay	18 k/bay	18 k/bay

Note!!! Wind loads shown are based on a design wind speed of 110 mph. For other design wind speed, V, multiply wind loads above by $V^2/110^2$.

TYPICAL PIPERACK CONFIGURATION

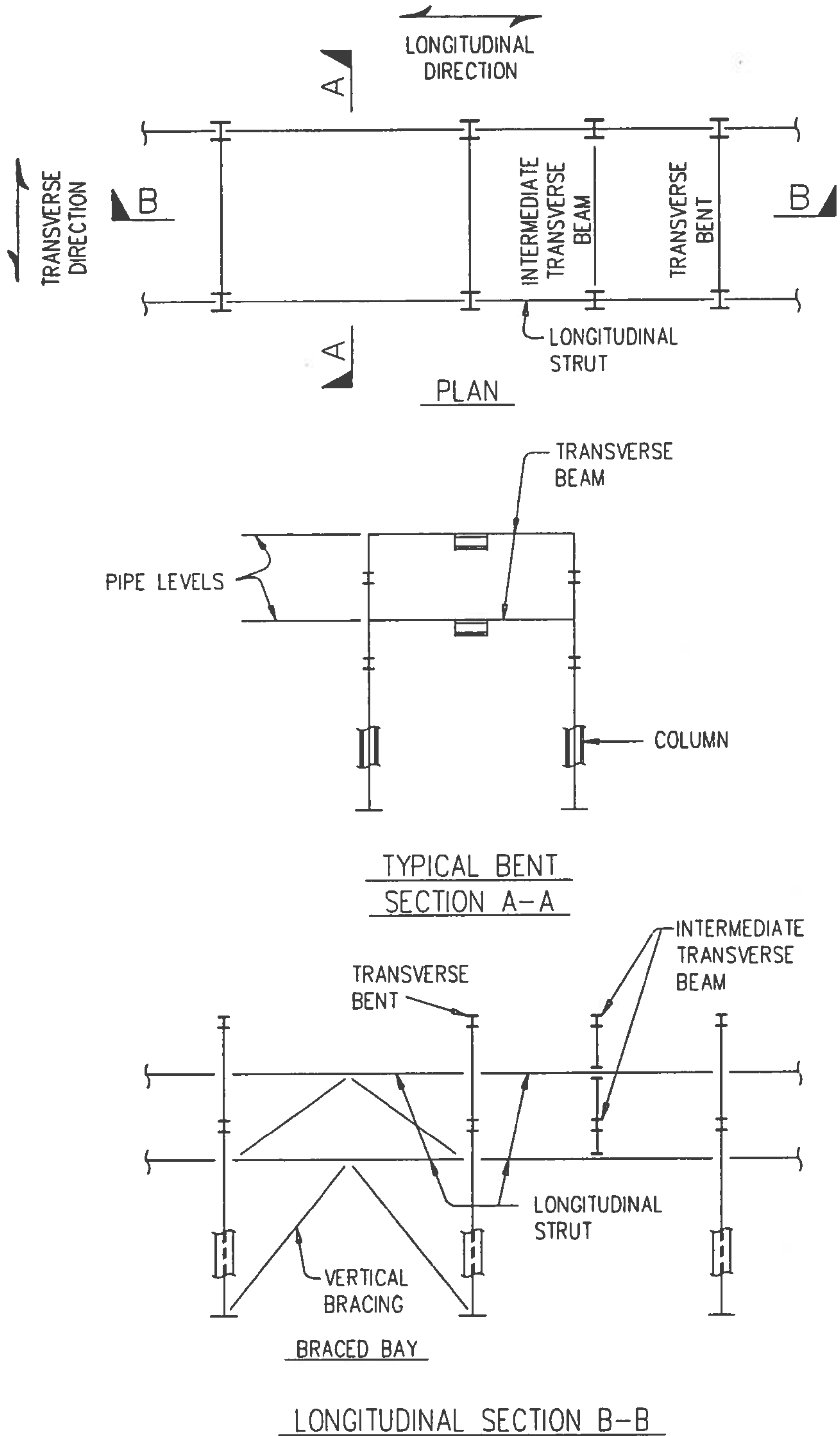


FIGURE 1: TYPICAL PIPE BRIDGE
FIGURE 2: MISCELLANEOUS PIPE SUPPORTS

Figure 1

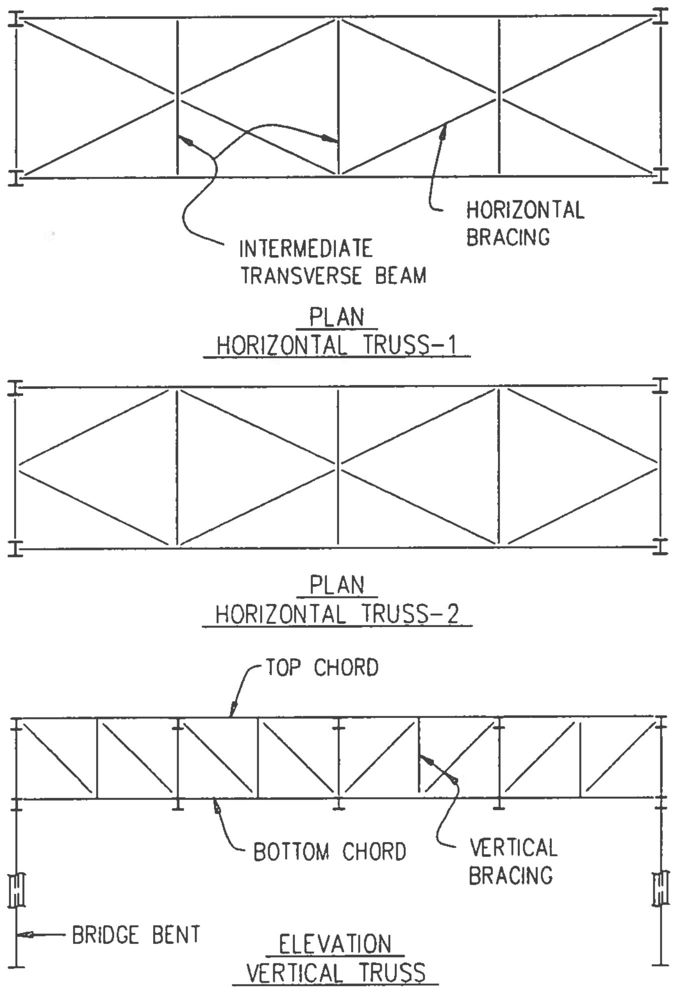
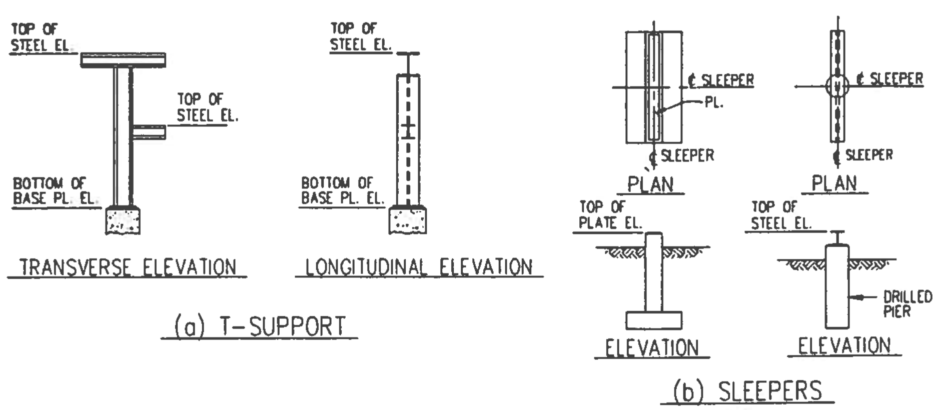
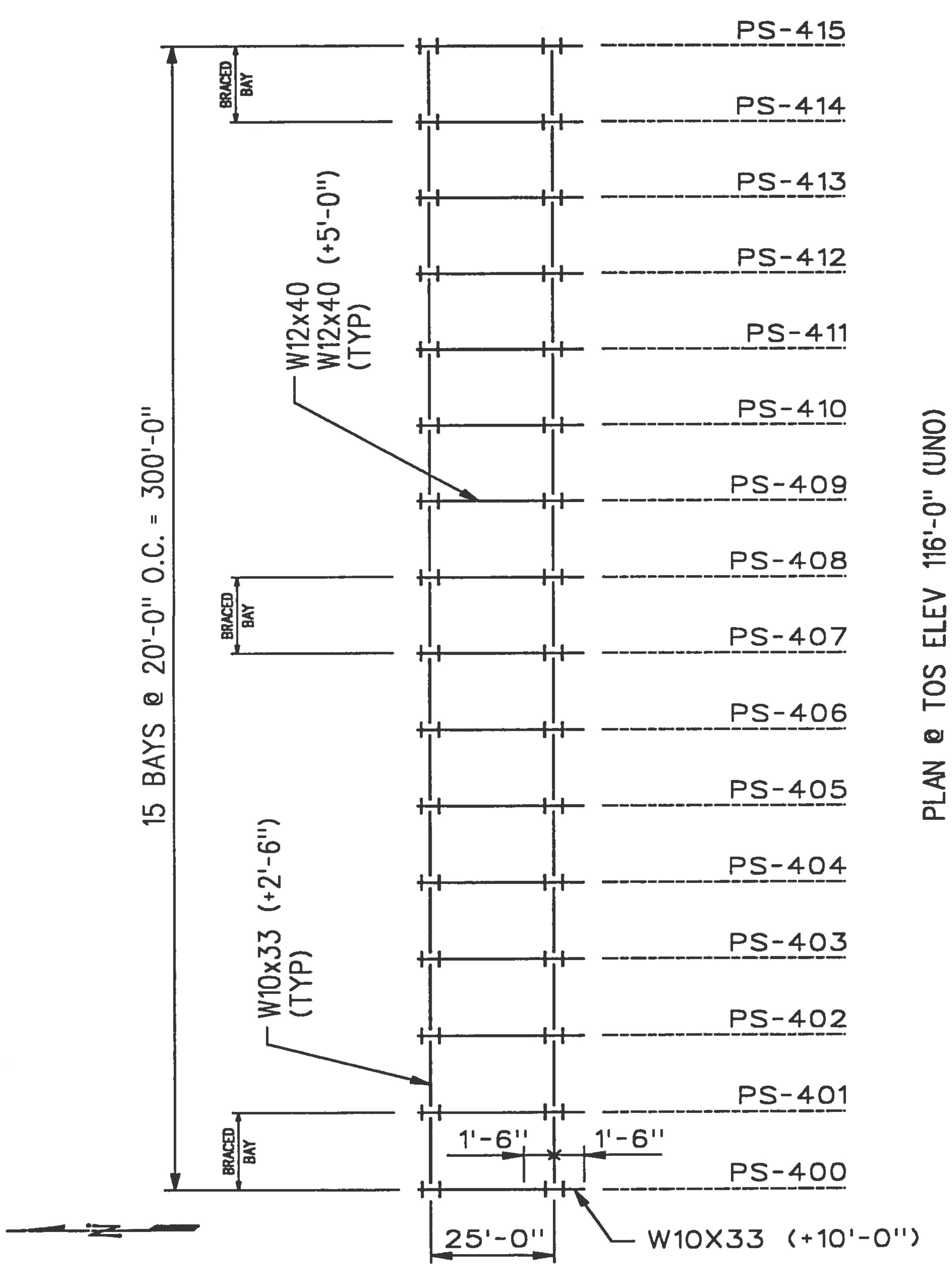


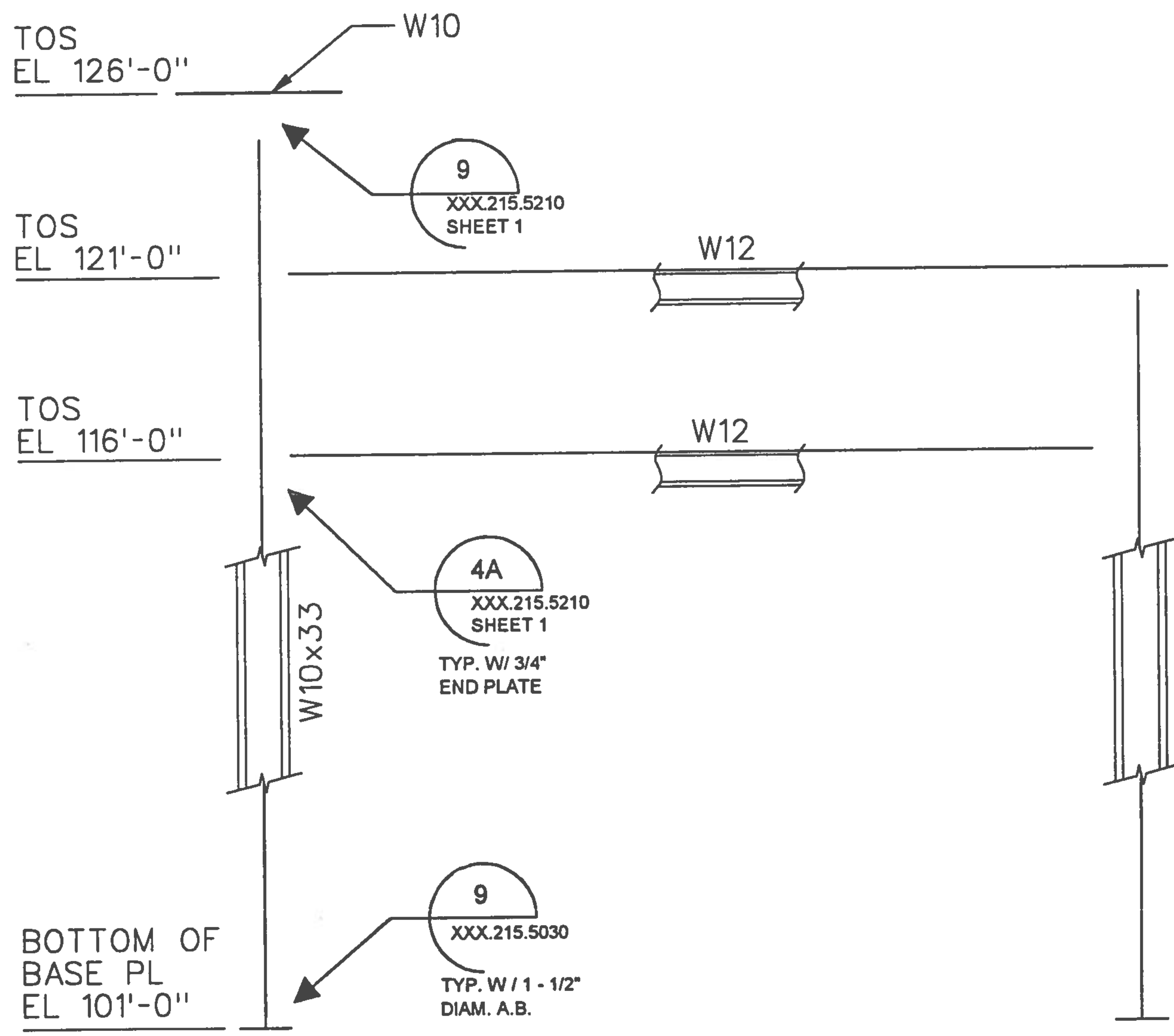
Figure 2



SAMPLE DESIGN 1 : STEEL PIPERACK DESIGN

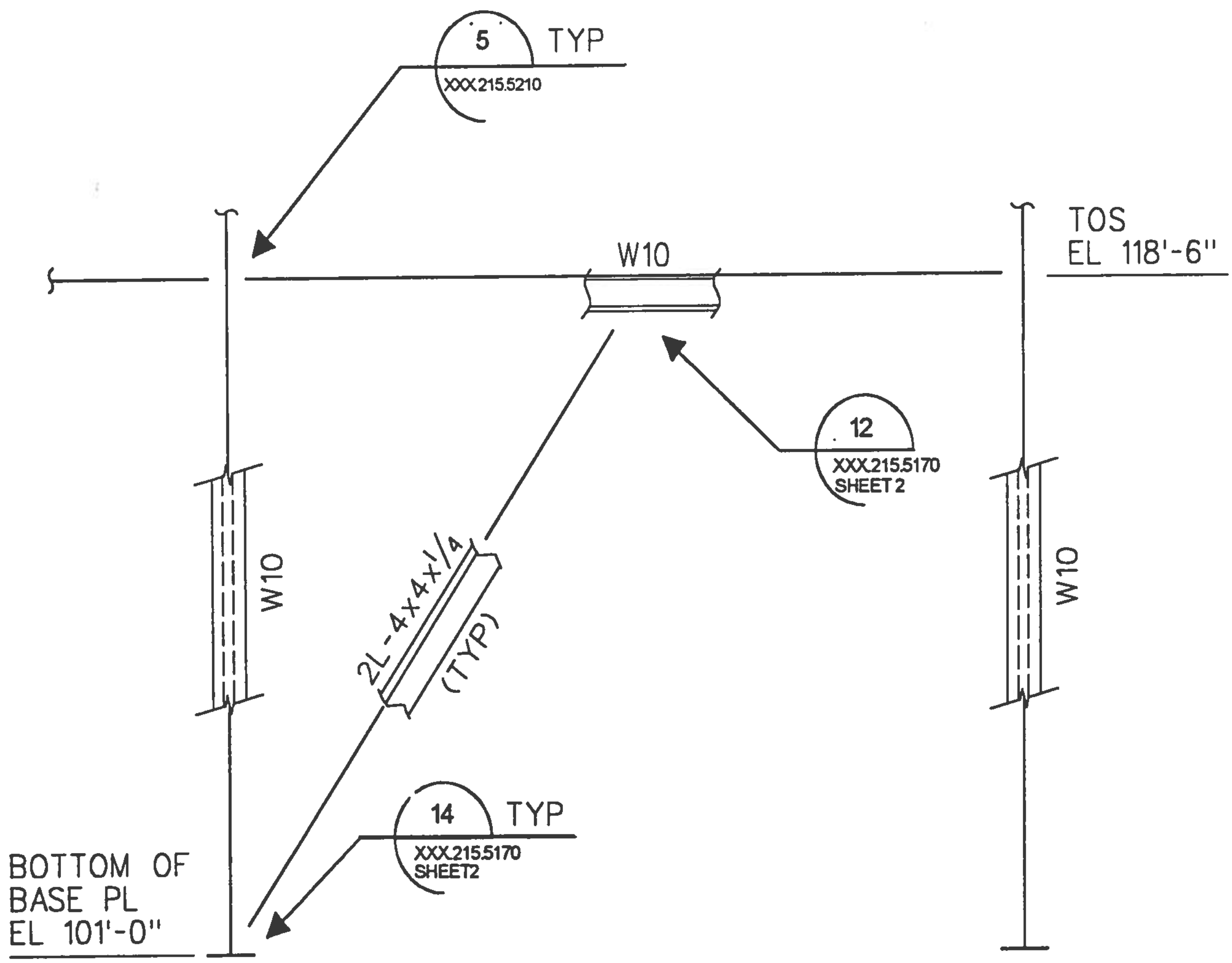


SAMPLE DESIGN 1 : STEEL PIPERACK DESIGN



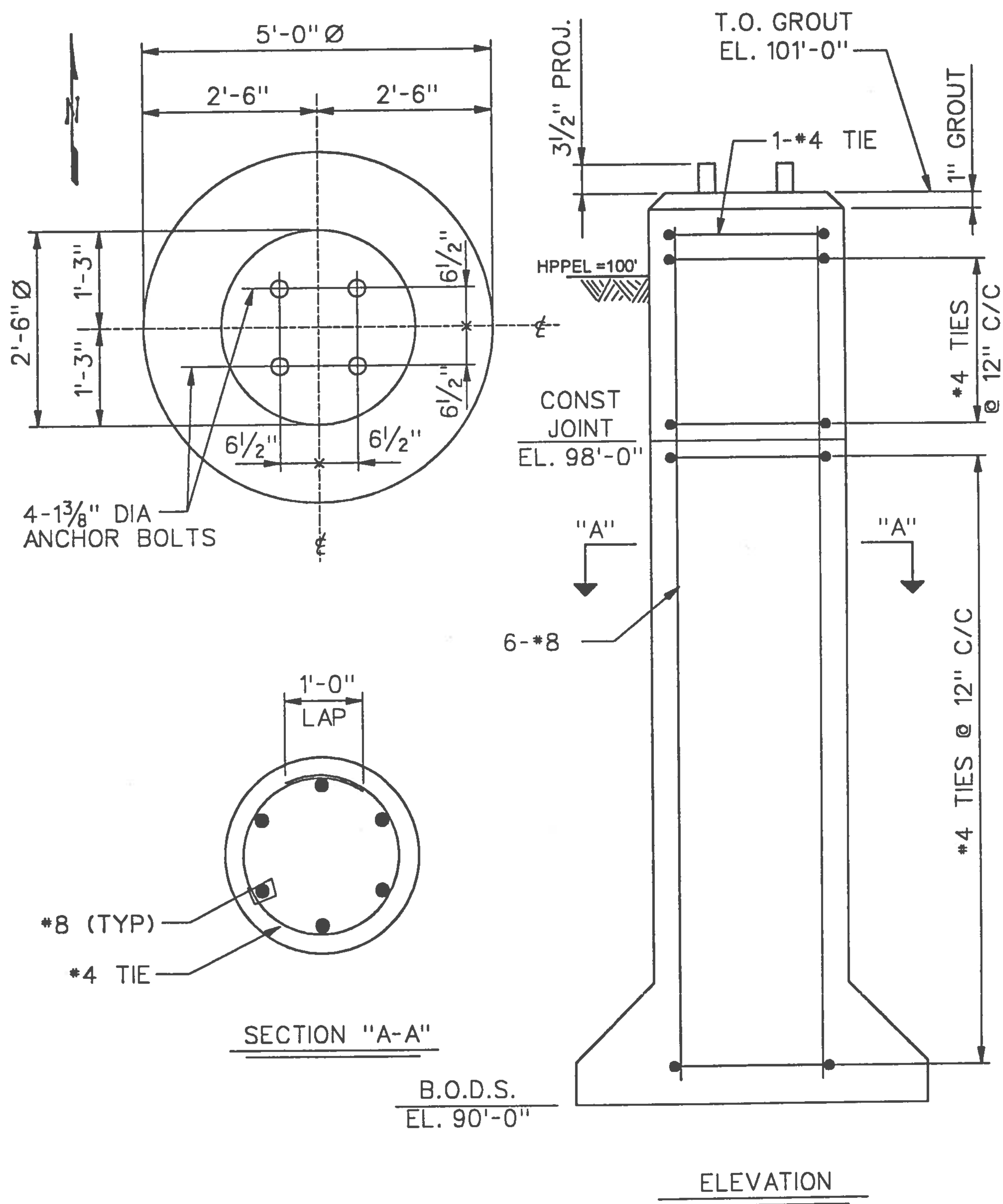
TYPICAL BENT ELEVATION

SAMPLE DESIGN 1 : STEEL PIPERACK DESIGN



TYPICAL BRACED BAY ELEVATION

SAMPLE DESIGN 1 : STEEL PIPERACK DESIGN



SAMPLE DESIGN 1 : STEEL PIPERACK DESIGN

GIVEN

References

- AISC Manual of Steel Construction, ASD 9th Edition
- Piping Drawings

Materials

- Steel - ASTM A36 (Allow 1/3 increase in allowable stresses for wind.)
- Bolts - ASTM A325N

Design Loads (20' Bay Spacing)

Gravity Loads:

Piping on beams @ TOS EL's 116'-0" & 121'-0" (Operating)

- O.L. = 0.04 ^{ksf} x 20' = 0.8 $^{K/ft}$ on members 6 & 7 - $0 \leq x \leq 18.5'$

24" ϕ Cooling water lines (@ TOS EL 121'-0")

- O.L. = 20(278.7 - 40 x 2) = 3.97^K on member 7 @ x = 20', 22.5'

24" ϕ Flare line (Assume 20% H_2O weight)

- O.L. = 20(94.7 + 0.2 x 184.0) = 2.63^K @ Joint 4

Note : Flare line will NOT be hydrotested on the piperack.

Transverse Wind:

- Wind on Piping and Struts, Joint Loads: 3.4^K @ JT.2, 5.6^K @ JT.3, 1.3^K @ JT.4
- Wind on Columns, Member Loads: 0.07^{KLF} on members 1 thru 3, 11, 12.

Note: Wind load calculations not shown for brevity. See practice 670.215.1215 "Wind Load Calculations" for procedures.

Load Combinations

- Basic Load Cases
  1. Gravity (Dead load of structure without piping)
  2. Piping (Weight of piping and contents)
  3. Wind Load (Transverse direction)

- Combinations
  4. Empty Condition = Gravity + 60% Piping
  5. Operating Condition = Gravity + Piping
  6. Empty + Wind = Empty Conditon + Wind Load
  7. Empty - Wind = Empty Condition - Wind Load
  8. Operating + Wind = Operating Condition + Wind Load
  9. Operating - Wind = Operating Condition - Wind Load

Note: Allow 1/3 increase in allowable stresses (0.75 factor in computer run) for combinations with wind.

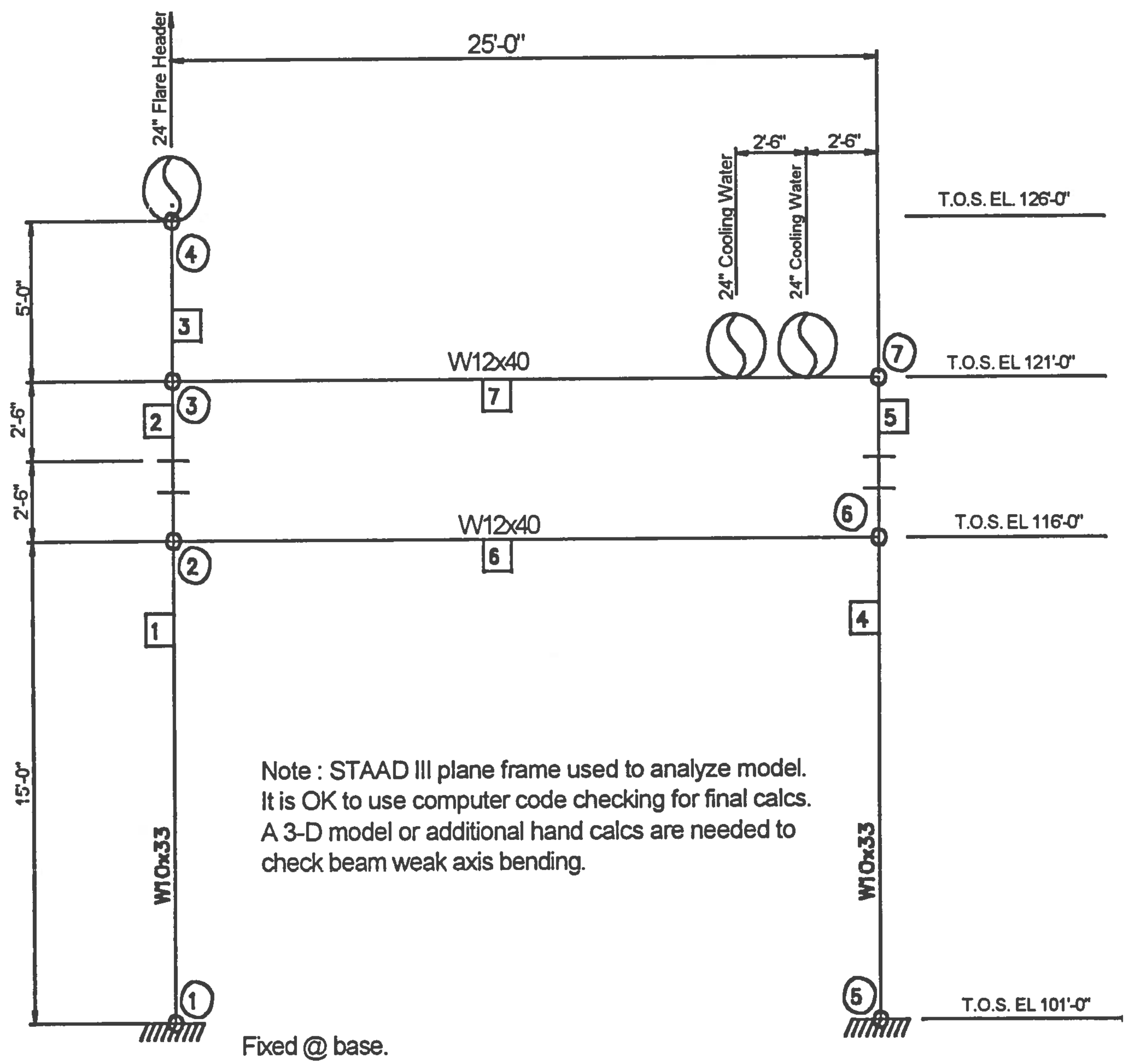
SAMPLE DESIGN 1 : STEEL PIPERACK DESIGN

REQUIRED

Determine steel member sizes, connections, and the foundations for the given loading conditions.

SOLUTION

Design Model



SAMPLE DESIGN 1 : STEEL PIPERACK DESIGN

Member Design

BOTTOM BEAM: MEMBER 6

Governing Load Combination: 9 (Operating - Wind), $M_x = 76.9^{Ft\text{-}K}$, $P = 15.1^K$ @ Joint #2

Try a W12x40 Beam

$$\left(\frac{KL}{r}\right)_y = \frac{1.0\ (2.5)\ (12)}{1.93} = 155.4 > C_c = 126.1 \Rightarrow F_a = 6.19\ \text{KSI}$$

$$f_a = \frac{15.1}{11.8} = 1.28\ \text{KSI}, \quad \frac{f_a}{F_a} = \frac{1.28}{6.19} = 0.21 > 0.15$$

$$\left(\frac{KL}{r}\right)_x = \frac{0.65\ (25)\ (12)}{5.13} = 38.0 \Rightarrow F'_{ex} = 103.42\ \text{KSI}$$

$$l_u = \frac{2.5}{3} = 8.33', \quad F_{bx} = 24\ \text{KSI} \qquad C_{mx} = 0.85$$

$$f_{bx} = \frac{76.9\ (12)}{51.9} = 17.78\ \text{KSI} \quad , \quad \frac{f_{bx}}{F_{bx}} = \frac{17.78}{24} = 0.74$$

$$\frac{f_a}{F_a} + \frac{C_{mx}\, f_{bx}}{\left(1 - \frac{f_a}{F'_{ex}}\right) F_{bx}} = 0.21 + \frac{0.85\ (0.74)}{\left(1 - \frac{1.28}{103.42}\right)} = 0.85 < 1.33 \quad \text{O.K.}$$

$$\frac{f_a}{0.6F_y} = \frac{f_{bx}}{F_{bx}} = \frac{1.28}{22} + 0.74 = 0.80 < 1.33 \quad \text{O.K.}$$

Use W12 x 40

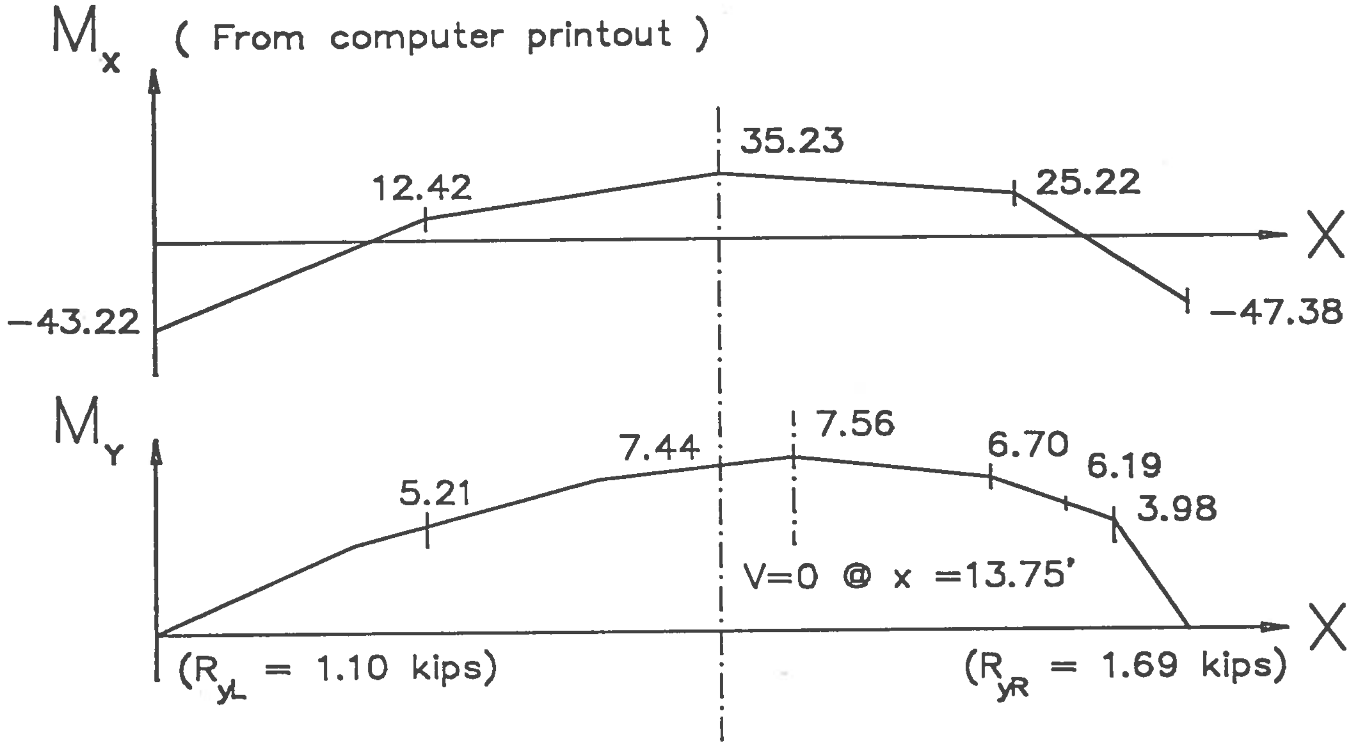
TOP BEAM: MEMBER 7

Governing Load Combination: 8 (Operating + Wind), $M_x = 68.2^{Ft\text{-}K}$, $P = 19.2^K$ @ JT. 7

Since loads are close to those for the bottom beam, Try a W12x40 beam

Check weak axis bending using operating load comb. 5, $P = 14.9^K$ + Friction Loads

SAMPLE DESIGN 1 : STEEL PIPERACK DESIGN



$f_a = \dfrac{14.9}{11.8} = 1.26$ KSI, $\dfrac{f_a}{F_a} = \dfrac{1.26}{6.19} = 0.20 > 0.15$

$f_{bx} = \dfrac{35.2(12)}{51.9} = 8.14$ KSI, $\dfrac{f_{bx}}{F_{bx}} = \dfrac{8.14}{24} = 0.34$

$f_{by} = \dfrac{7.6(12)}{11.0} = 8.29$ KSI, $\dfrac{f_{by}}{F_{by}} = \dfrac{8.29}{27} = 0.31$

$$\frac{f_a}{F_a} + \frac{C_{mx} f_{bx}}{\left(1 - \frac{f_a}{F'_{ex}}\right) F_{bx}} + \frac{C_{my} f_{by}}{\left(1 - \frac{f_a}{F'_{ey}}\right) F_{by}} = 0.20 + \frac{0.85}{\left(1 - \frac{1.26}{103.42}\right)}(0.34) + \frac{1.0}{\left(1 - \frac{1.26}{6.19}\right)}(0.31)$$

$$= 0.20 + 0.29 + 0.39 = 0.88 < 1.0 \quad \text{O.K.}$$

$\boxed{\text{Use W12 x 40}}$

SAMPLE DESIGN 1 : STEEL PIPERACK DESIGN

COLUMNS

Member 5, Load Comb. 8 (Operating + Wind), $M_x = 68.2^{Ft\text{-}K}$, $P = 19.2^K$ @ Joint 7
Try W10x33 Column

$$\left(\frac{KL}{r}\right)_y = \frac{1.0(17.5)(12)}{1.94} = 108.2 < 126.1$$

$$\frac{KL/r}{C_c} = \frac{108.2}{126.1} = 0.86$$

$$F_a = C_a F_y = 0.33 \times 36 = 11.88 \text{ KSI}$$

$$f_a = \frac{19.2}{9.71} = 1.98 \text{ KSI}, \quad \frac{f_a}{F_a} = \frac{1.98}{11.88} = 0.17 > 0.15$$

$$f_{bx} = \frac{68.2(12)}{35.0} = 23.38 \text{ KSI}, \quad \frac{f_{bx}}{F_{bx}} = \frac{23.38}{22} = 1.06$$

From Figure C-C2.2, AISC Manual,

$$G_{Ax} = \frac{(170/15 + 170/5)}{(310/25)} = 3.66, \quad G_{Bx} = \frac{(170/5)}{(310/25)} = 2.74, \quad K_X \approx 1.85$$

$$\left(\frac{KL}{r}\right)_x = \frac{1.85(5)(12)}{4.19} = 26.5, \quad F'_{ex} = 229.92 \text{ KSI}$$

$$\frac{f_a}{F_a} + \frac{C_{mx} f_{bx}}{\left(1 - \frac{f_a}{F'_{ex}}\right) F_{bx}} = 0.17 + \frac{0.85}{\left(1 - \frac{1.98}{229.92}\right)}(1.06) = 1.08 < 1.33 \quad \text{O.K.}$$

$$\frac{f_a}{0.6F_y} + \frac{f_{bx}}{F_{bx}} = \frac{1.98}{22} + 1.06 = 1.15 < 1.33 \quad \text{O.K.} \quad \Leftarrow$$

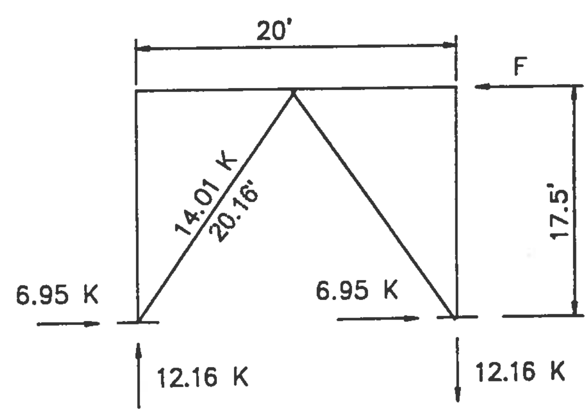
$$\boxed{\text{Use W10x33 Column}}$$

BRACED BAY

Friction Force:

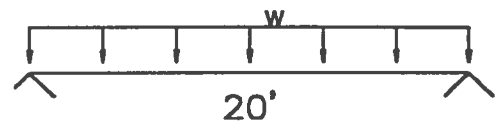
Oper. Shear on Trans. Beams = 10.36^K

$$F = 0.1 \times (10.36 + 15.63) \times \frac{16 \text{ Bents}}{3 \text{ Braced Bays}} = 13.9^{K} / \text{Side}$$



SAMPLE DESIGN 1 : STEEL PIPERACK DESIGN

Strut:
Heaviest Trans Beam Shear = 15.64^K



$$w = \frac{15.64}{(25/2)} \times 0.5 = 0.63 \text{ KLF}$$

$$M = 0.63 \frac{(20)^2}{8} = 31.5^{Ft-K}$$

From Beam Tables in AISC, p2-173 , Try W10x33 , $M_{Allow} = 52^{Ft-K}$

From Column Tables, AISC, p3-30, $P_{Allow} = 95^K$

$$\sum \frac{f_i}{F_i} = \frac{13.9}{95} + \frac{31.5}{52} = 0.15 + 0.61 = 0.76 < 1.0 \quad \text{O.K.}$$

Use W10x33 Strut

BRACING

$P_{MAX} = 14.0^K$, $(KL)_y = (KL)_x = 20.2'$

From Column Tables, Use 2L4 x 4 x 1/4 ($P_{Allow} = 16^K$)

$$\left(\frac{KL}{r}\right)_{X\ min} = \frac{20.2(12)}{1.25} = 194 < 200 \quad \text{O.K.}$$

Use 2- L4 x 4 x 1/4 for Bracing

Connections

MOMENT CONNECTION

Note: Moment Connections not shown for brevity. See technical practice 670.215.1209 " Bolted End Plate Moment Connections" for procedures.

Use 4A XXX.215.5210, Sheet 1 (3/4" End PL w/ 5/16" welds)

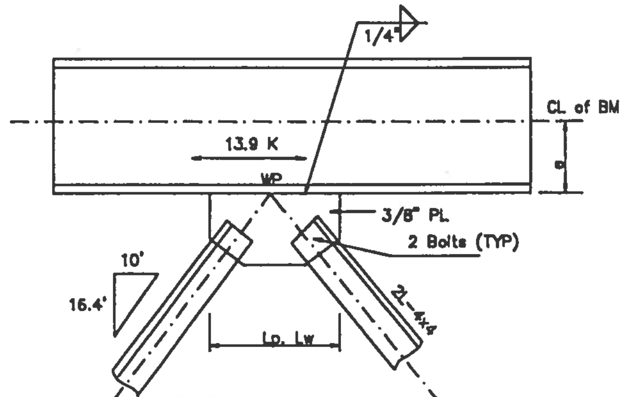
BASE PLATE

Note: Base Plate design not shown for brevity. See technical practice 670.215.1208 " Base Plate Design Criteria " for procedures. Also see practice 670.215.1207, Anchor Bolt Design Criteria, for anchor bolt design procedure.

Use STD. Base PL detail 9 / XXX.215.5030 for W10 column, with 1-1/2 " ϕ Anchor Bolts

SAMPLE DESIGN 1 : STEEL PIPERACK DESIGN

BRACING CONNECTIONS



DETAIL 12 / XXX.215.5170, Sheet 2
(STD.)

- Bolts : A - 325 - N - 3/4" ϕ

Min Allow = 18.6^K , Tables 1-D & 1-E, AISC, p4-5,6

Bolts REQ'D = 14/18.6 = 1

Use 2 - 3/4" ϕ A-325-N Bolts

- Gusset PL : 3/8" PL per STD.
  F_V = 14.4 KSI

$$L_P \text{ REQ'D} = \frac{13.9}{(0.375)(14.4)} = 2.6"$$

- Weld : 1/4" per STD.

$$L_W \text{ REQ'D} = \frac{V}{0.928\ D \times (2\ \text{Sides})} = \frac{13.9^K}{0.928\ (4)\ 2} = 1.87"$$

Use STD. DET. 12 / XXX.215.5170, Sheet 2

Note: Where WT bracing is used w/ flange attached to the gusset PL, it is required that the eccentric moment due to the offset between the centerline of the column / beam and the centroid of the WT be considered in the design of the WT section.

- Gusset PL : 3/8" PL per STD.

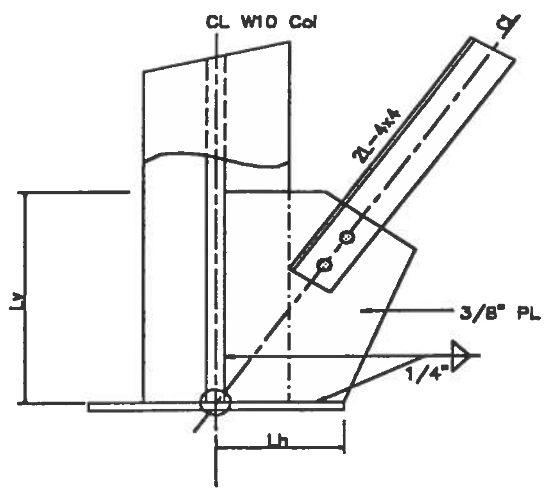
$$L_{VP} \text{ REQ'D} = \frac{12.19}{(0.375)(14.4)} = 2.26"$$

$$L_{HP} \text{ REQ'D} = \frac{6.95}{(0.375)(14.4)} = 1.29"$$

- Weld : 1/4" per STD.

$$L_{VW} \text{ REQ'D} = \frac{12.19}{(0.928)(4)2} = 1.64$$

$$L_{HW} \text{ REQ'D} = \frac{6.95}{(0.928)(4)2} = 0.94"$$



DETAIL 14 / XXX.215.5170, Sheet 2
(STD.)

SAMPLE DESIGN 1 : STEEL PIPERACK DESIGN

- Check STD shear clip angle connection for tension on strut, $T_{MAX} = 13.9^K$
  Per AISC Hanger - Type connections, p4-89

  For 3/8" x 5-1/2" LG clip angles , b = 2, $T_{Allow} = 5.5 \times 1.27 = 7.0^K < 13.9^K$ N.G.

∴ Clip angles are inadequate

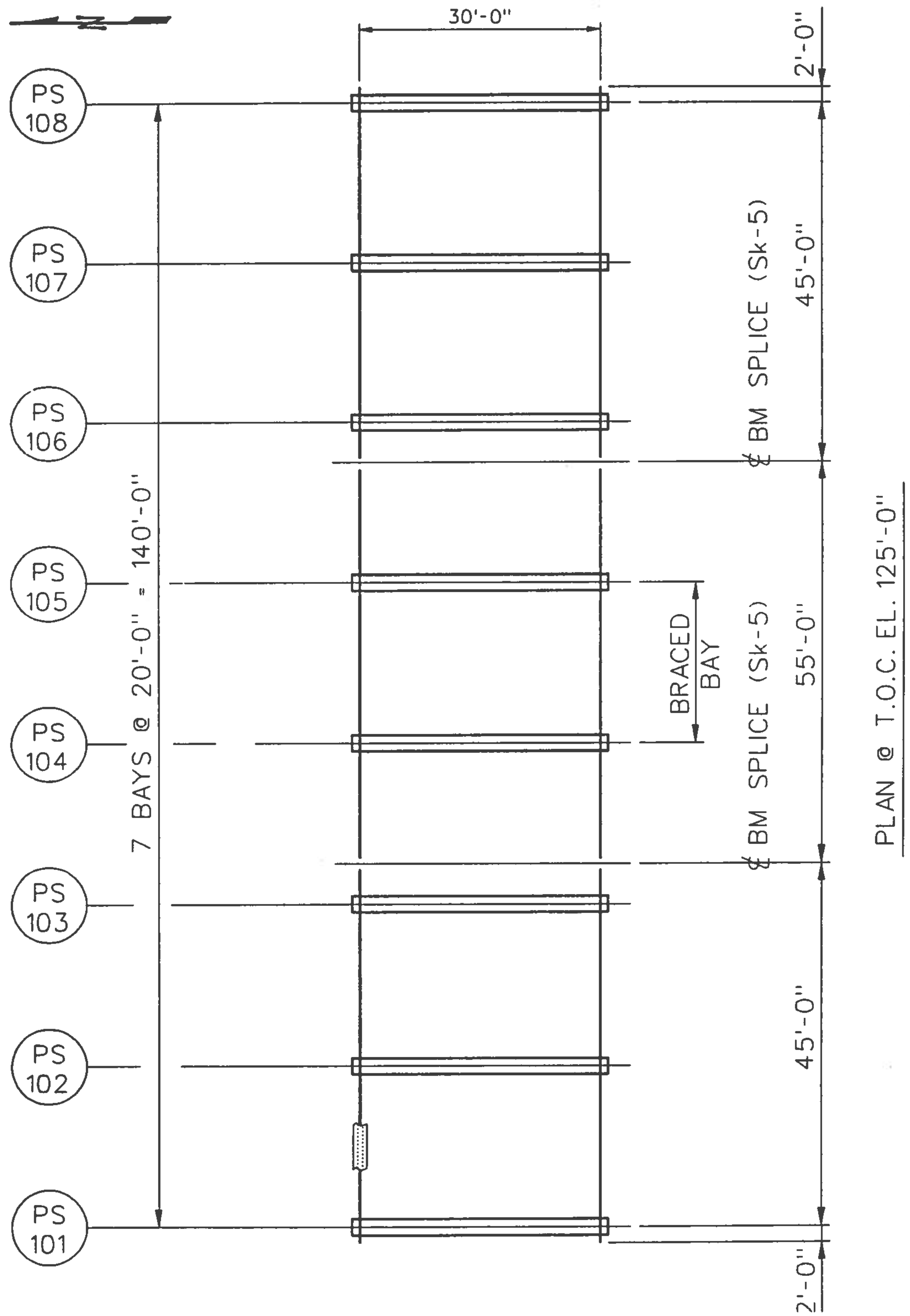
Use STD Shear PL Connection for Struts to Columns

<u>Foundation Design</u>

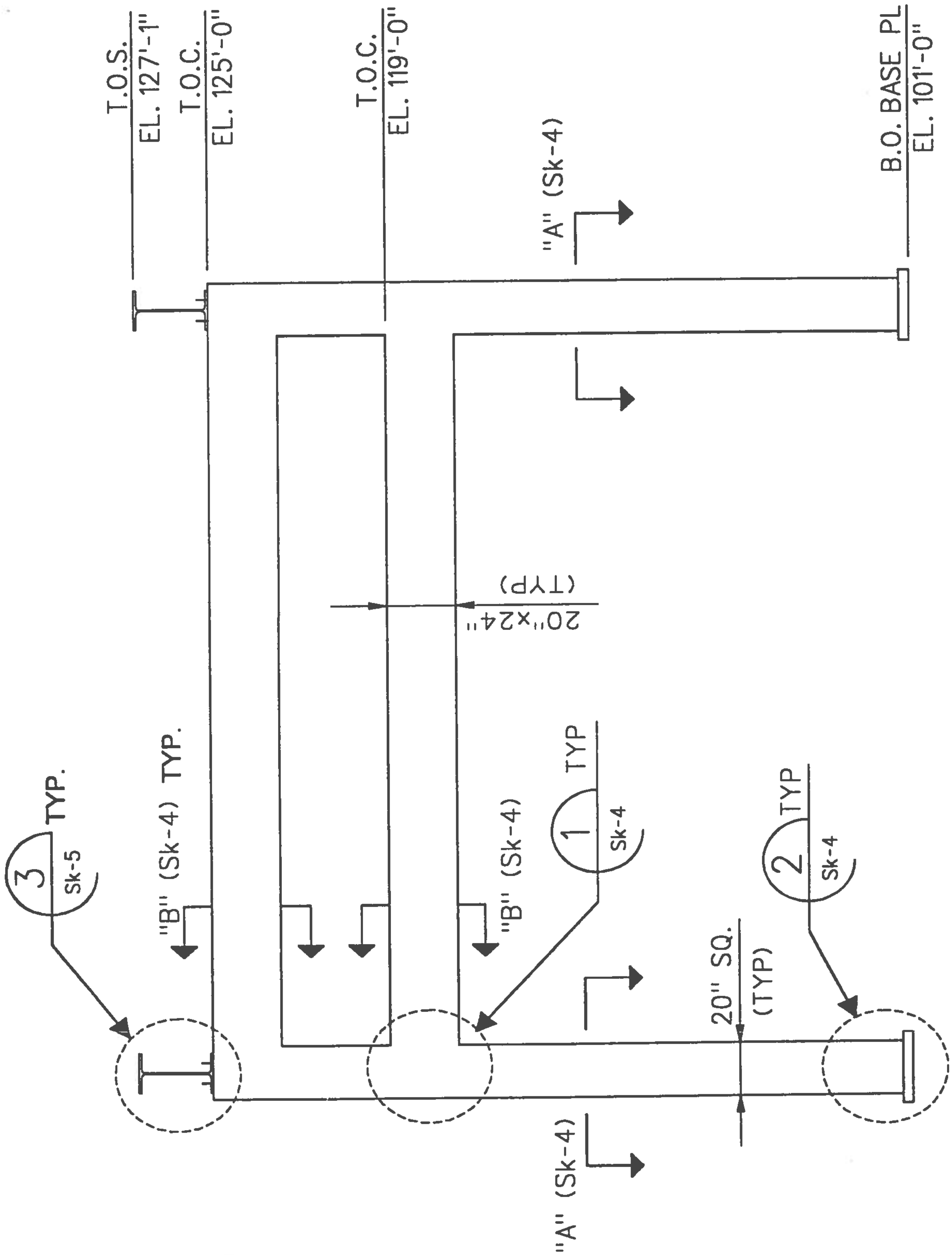
Note : Foundation design not shown for brevity. See technical practice 670.215.1231 for drilled pile foundations. See technical practice 670.215.1232 for driven pile supported foundations.

FLUOR DANIEL

SAMPLE DESIGN 2 : CONCRETE PIPERACK DESIGN

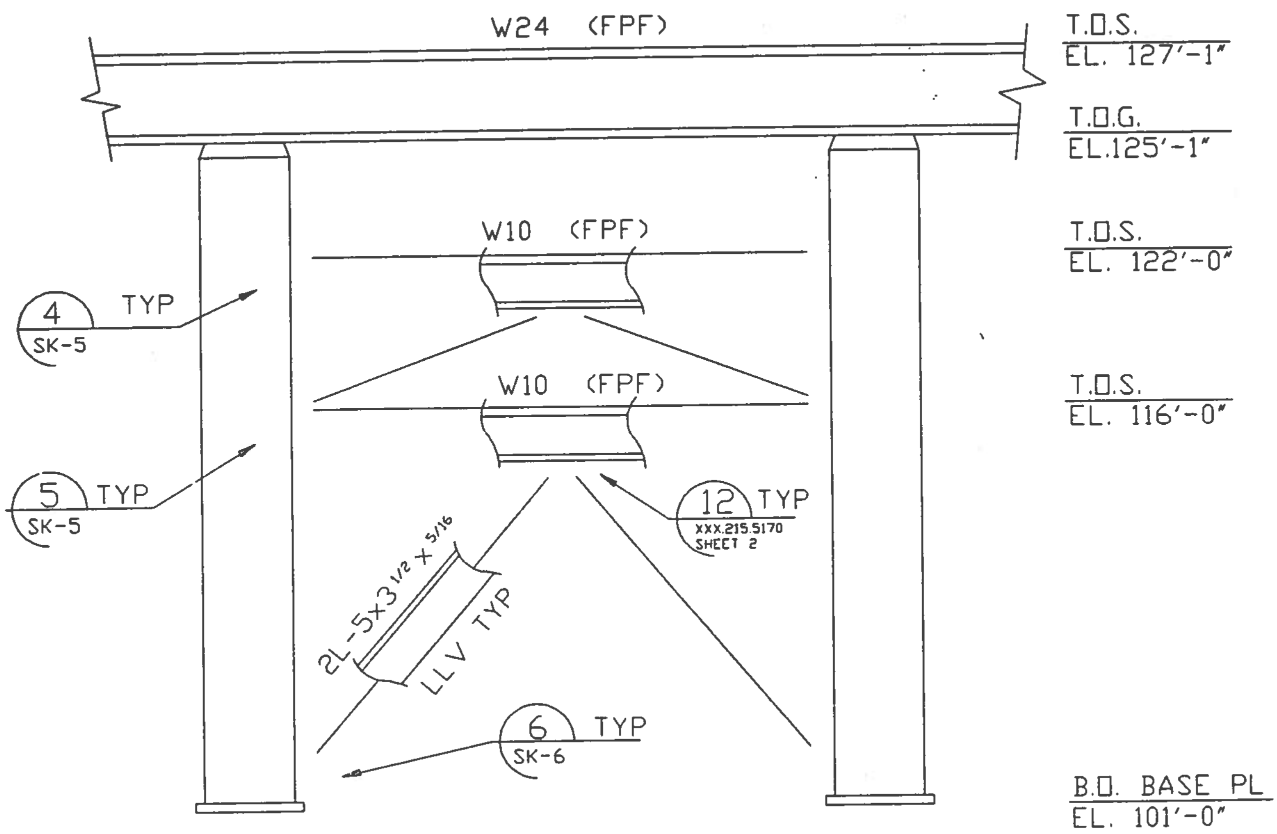


SAMPLE DESIGN 2 : CONCRETE PIPERACK DESIGN

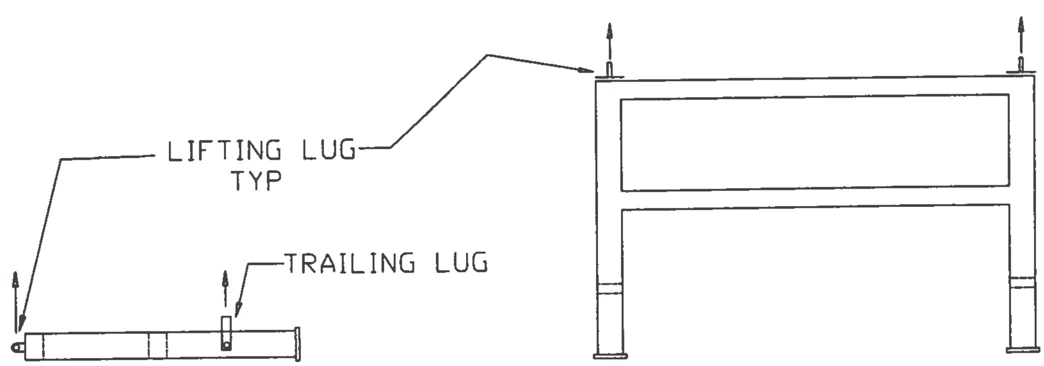


TYPICAL BENT ELEVATION

SAMPLE DESIGN 2 : CONCRETE PIPERACK DESIGN

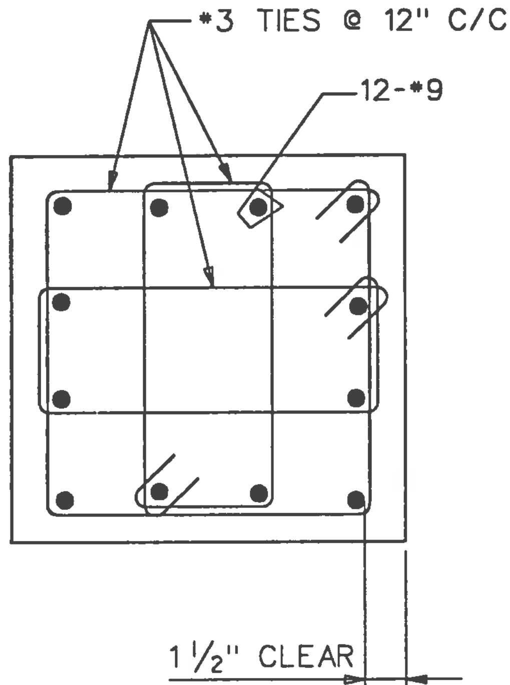


TYPICAL BENT ELEVATION

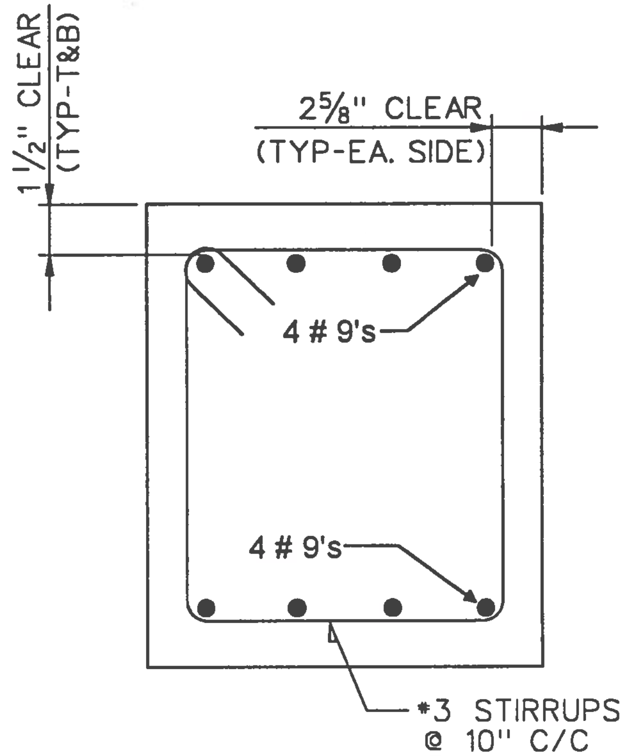


LIFTING ARRANGEMENT

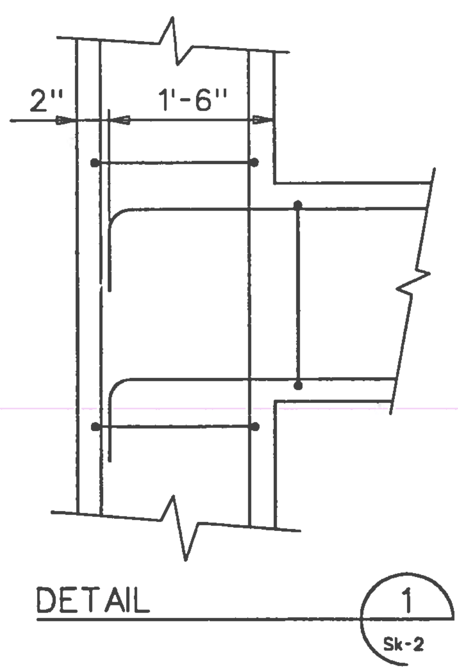
SAMPLE DESIGN 2 : CONCRETE PIPERACK DESIGN



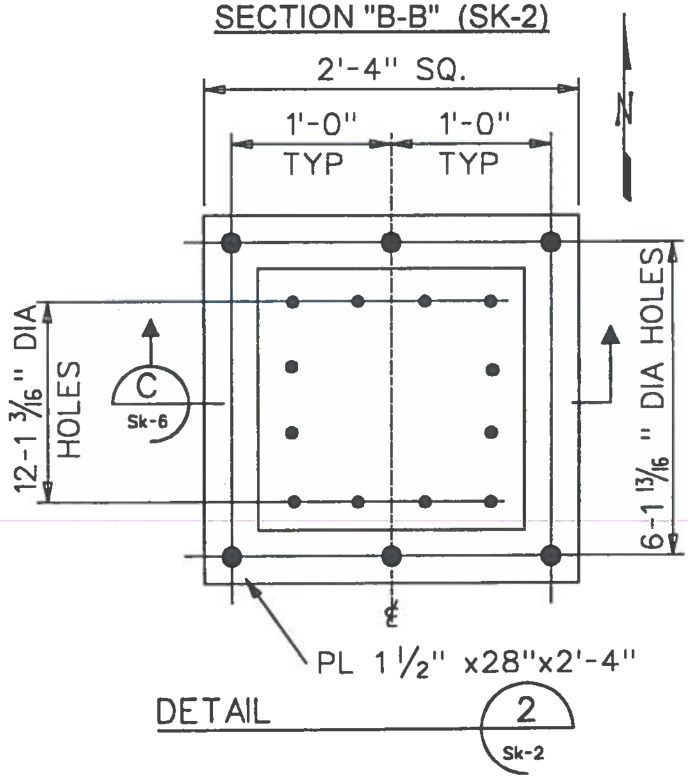
SECTION "A-A" (SK-2)



SECTION "B-B" (SK-2)

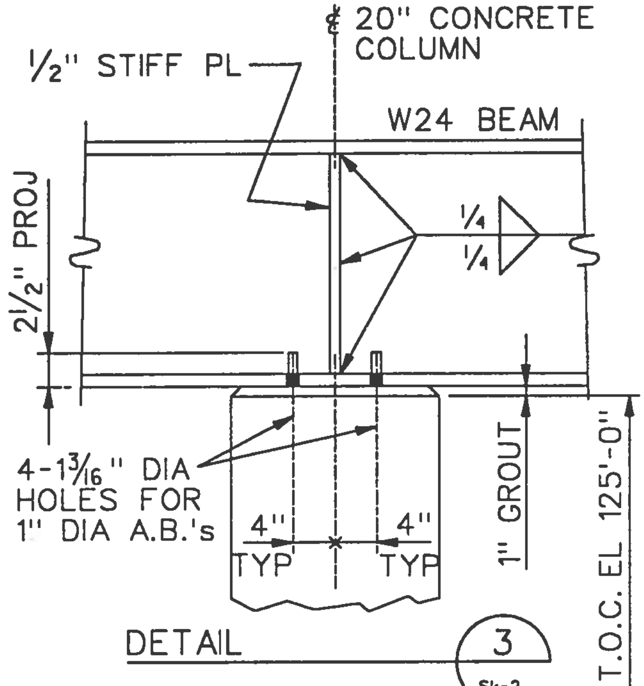


DETAIL

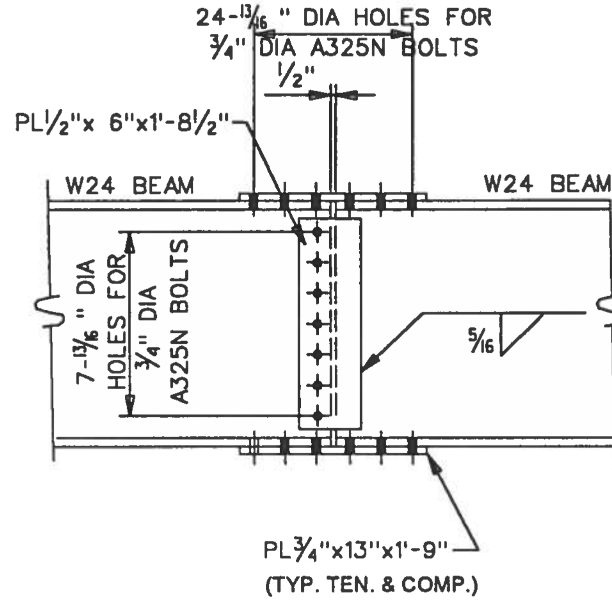


DETAIL

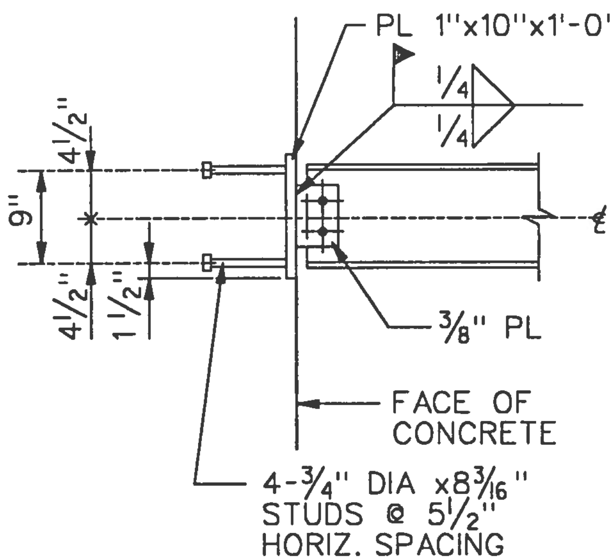
SAMPLE DESIGN 2 : CONCRETE PIPERACK DESIGN



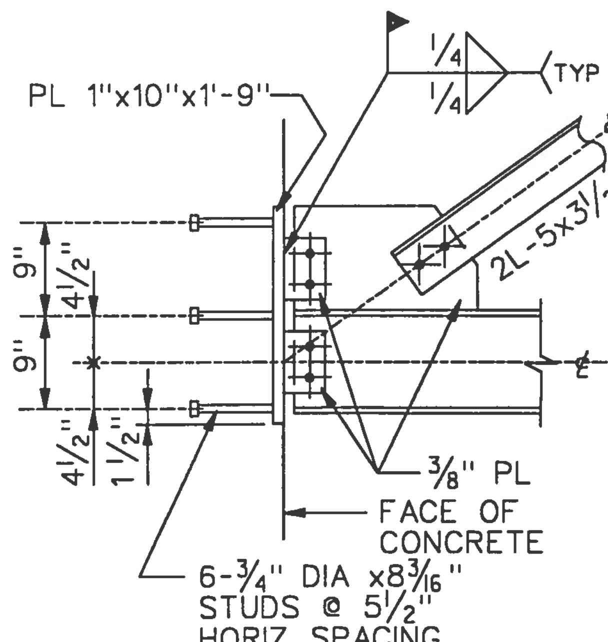
DETAIL 3



BEAM SPLICE DETAIL (Sk-1)

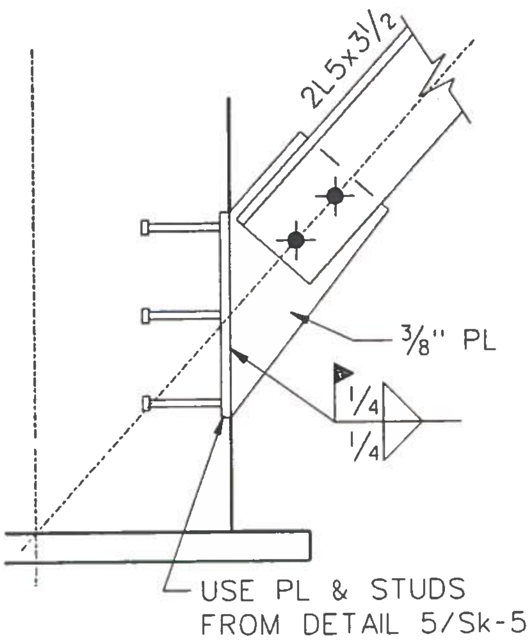


DETAIL 4

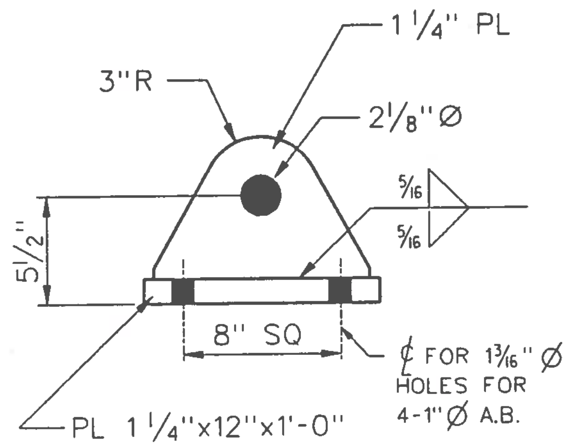


DETAIL 5

SAMPLE DESIGN 2 : CONCRETE PIPERACK DESIGN

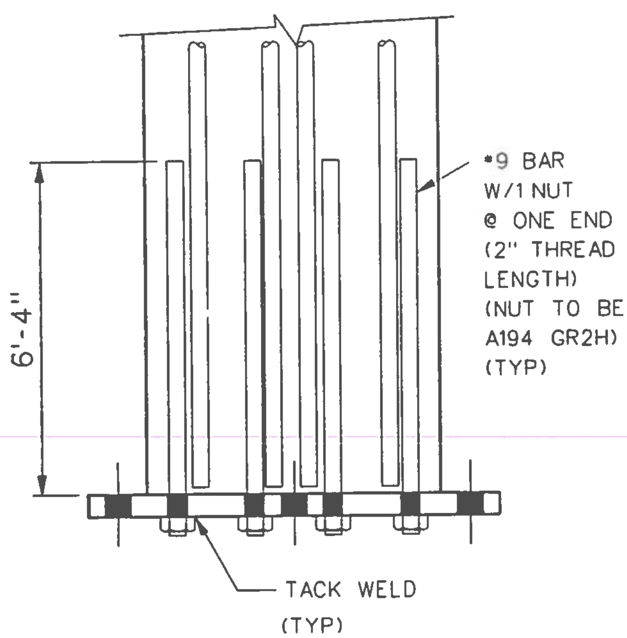


DETAIL 6 Sk-3

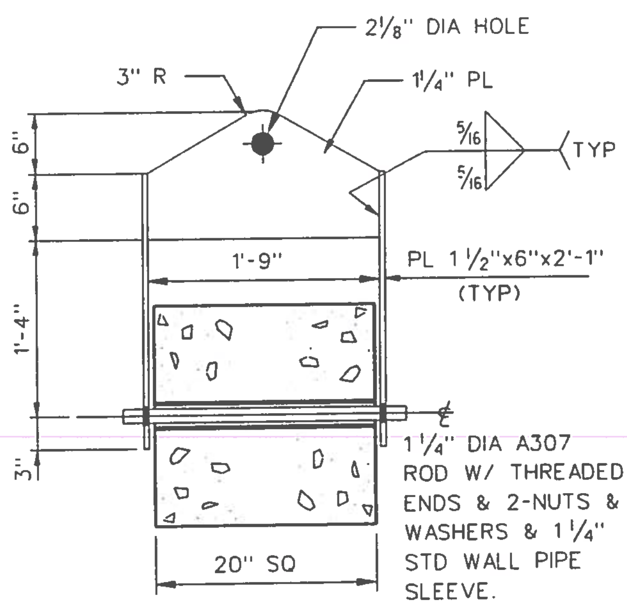


LIFTING LUG DETAIL

NOTE:
LIFTING ARRANGEMENT / DETAIL SHALL BE VERIFIED BY THE RIGGING CONTRACTOR.

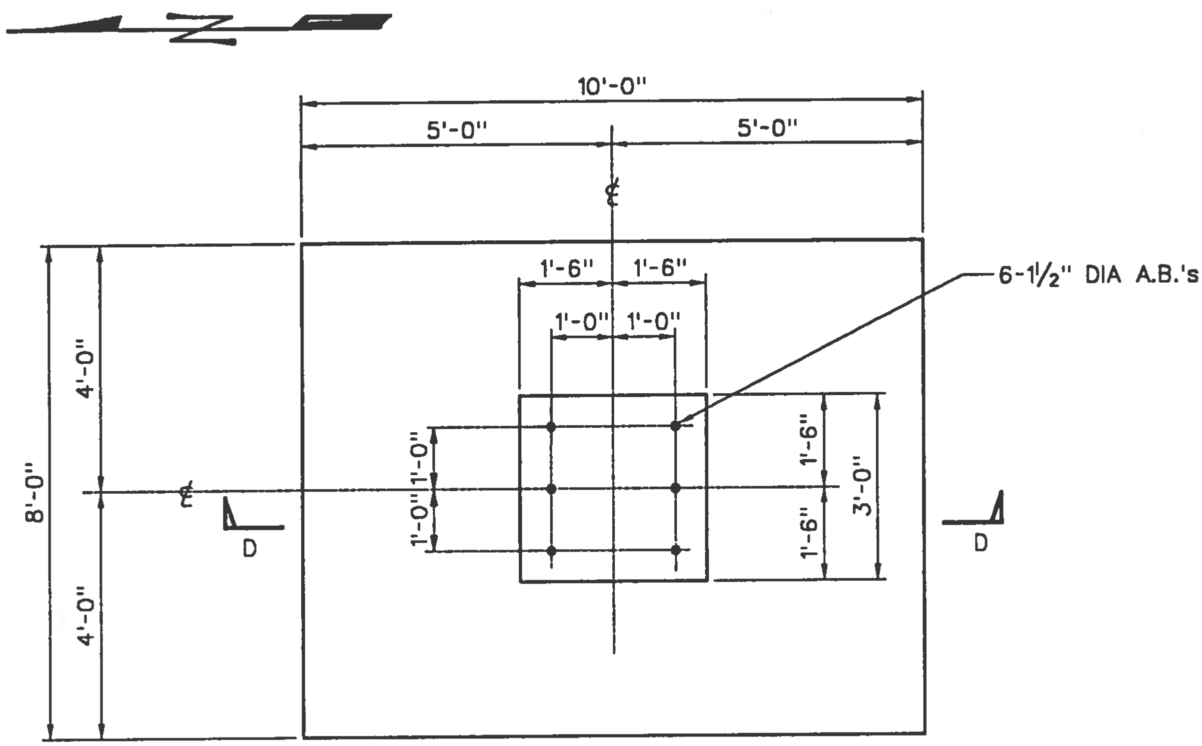


SECTION "C-C" (Sk-4)

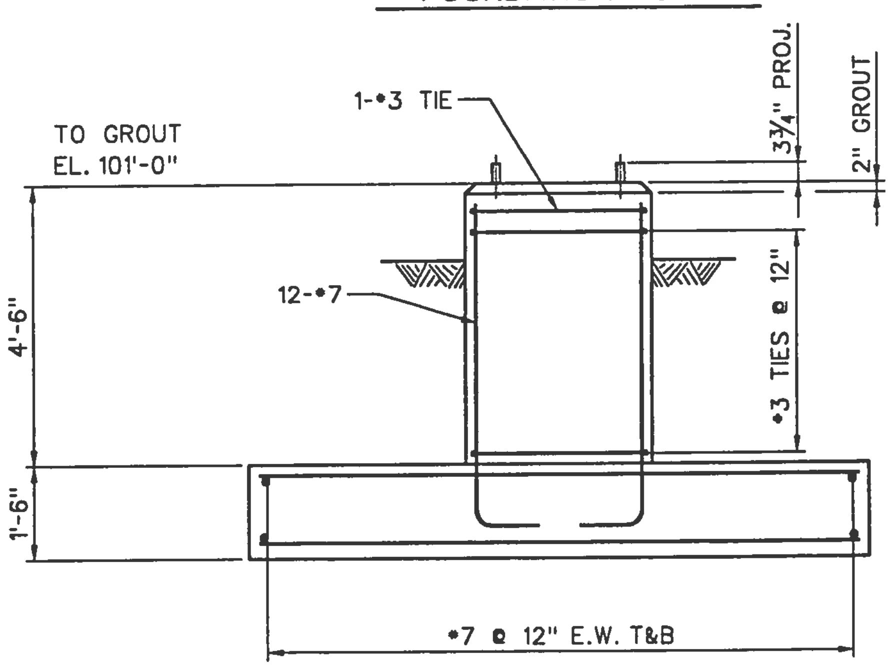


TAILING LUG DETAIL

SAMPLE DESIGN 2 : CONCRETE PIPERACK DESIGN



FOUNDATION PLAN



SECTION "D-D"

SAMPLE DESIGN 2 : CONCRETE PIPERACK DESIGN

GIVEN

References

- ACI- 318-95
- AISC Steel Manual -ASD 9th ed.
- Piping Plans
- Air Cooler Vendor Data

Materials

- Concrete : f_c' = 4000 PSI for P/R Bent, f_c' = 3000 PSI for Foundation, γ_c = 150 PCF
- Reinforcing Steel : f_y = 60 KSI
- Steel : f_y = 36KSI - Bolts : 3/4" ϕ A325N
- Anchor Bolts : A36
- Soil : Allow Net Soil Bearing = 3 KSF @ 5' Below grade. (Allow 1/3 increase due to wind)
  γ_s = 120 PCF
  Water Table Depth = 6' Below Grade

Design Loads

Gravity

- Structure : Include weight of Concrete and Steel members
- Piping (Operating) : w = 0.04 x 20' = 0.8 KLF @ member 5 - 6' $\le x \le$ 30' , and member 6.
  - 24" ϕ Cooling Water Lines : P = 20' (0.2787 - 2 (0.04)) = 3.97 ^{K} @ member 5, x = 2' , 4.5'
- Air Coolers : Vendor info not available, use air cooler loads from table No. 3 - Apply @ top of steel elev. 127'-1", @ JTS. 5 & 6 with eccentricity from shear applied @ top of steel beams.

Transverse Wind

Note : Wind Calculations not shown for brevity. See technical practice 670.215.1215 for procedures.

Joint Loads:
Fx : 2.9^K @ JTS. 3 & 4, 11.2^K @ JTS. 5 & 6 .
Mz : -12.5^{Ft-K} @ JTS. 5 & 6

Member Loads: (Wind on Columns) 0.12 KLF on members 1 thru 4.

Load Combinations

- Basic Loads (Note: Since Live Load on air coolers is small; multiply live load by (1.7/1.4) and include with piping.)
  1. Gravity Load without piping & air cooler live loads.
  2. Piping (Wt of piping and contents) & air cooler live loads.
  3. Transverse wind loads.
-
- Combinations (Unfactored comb's 4 thru 9 are the same as the factored comb's w/o load factors.)

SAMPLE DESIGN 2 : CONCRETE PIPERACK DESIGN

- Factored

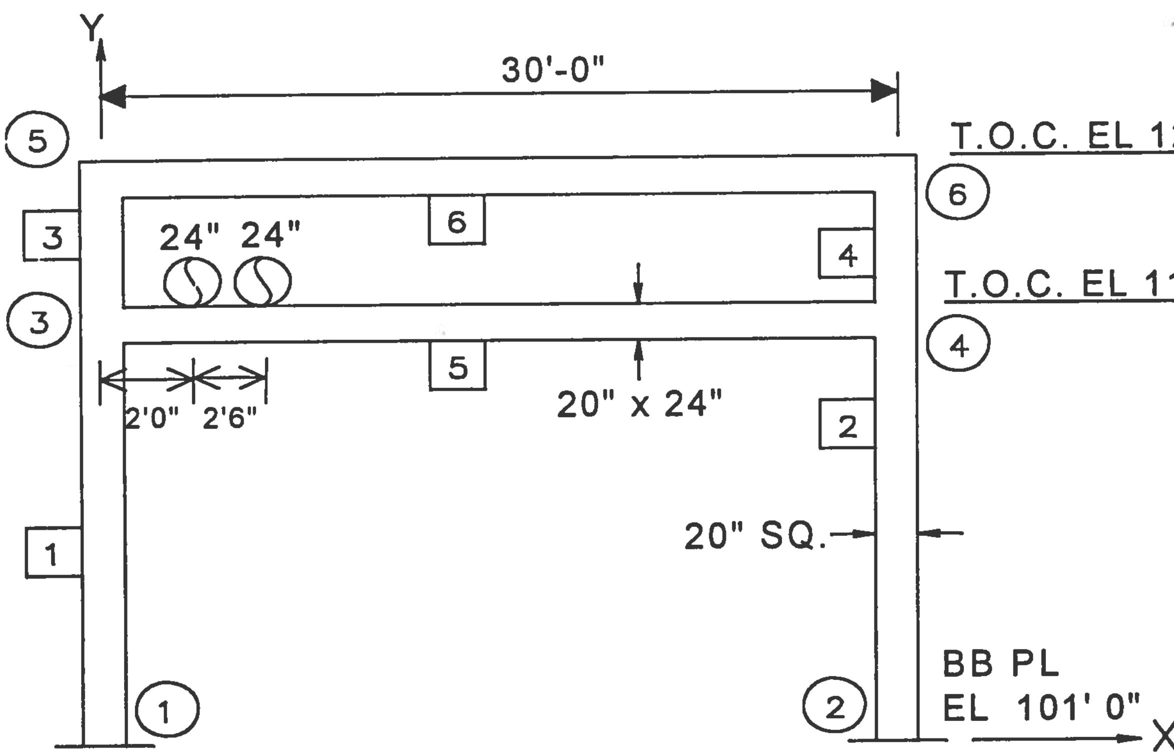
10. Empty = 1.4(Load 1 + 0.6 x Load 2)
11. Operating = 1.4 (Load 1 + Load 2)
12. Empty + Wind = 0.9 (Load 1 + 0.6 x Load 2) + 1.3 (Load 3)
13. Empty - Wind = 0.9 (Load 1 + 0.6 x Load 2) - 1.3 (Load 3)
14. Oper. + Wind = 0.75 {(Load 11) + 1.7 (Load 3)}
15. Oper - Wind = 0.75 {(Load 11) - 1.7 (Load 3)}

REQUIRED

Design concete members and detail connections for the given loading.

SOLUTION

Computer Model



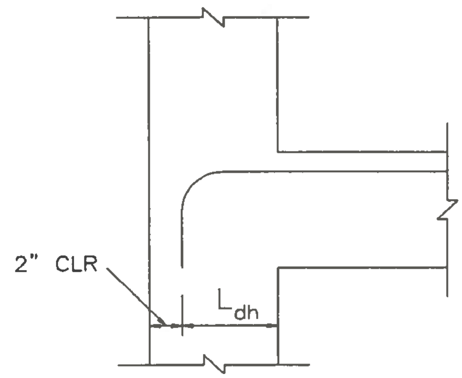
Note: STAAD III used to analyse model. It is okay to use the concrete capabilities of STAAD III; however, as of this time, STAAD III does not consider tension in columns, nor does it perform torsional analysis.

SAMPLE DESIGN 2 : CONCRETE PIPERACK DESIGN

Member Design

Beams : Try 20" wide x 24" deep beams. (d = 24 -2-1/2" = 21- 1/2")

Design Cases:
Member 5, Joint 3, Comb. 15, $Mu^- = 299'^K$
Member 5, Joint 4, Comb. 13, $Mu^+ = 135'^K$



$$F = \frac{bd^2}{12000} = \frac{20(21.5)^2}{12000} = 0.770$$

$$K_u^- = \frac{M}{F} = \frac{299}{0.77} = 388, \quad a_u = 4.20$$

$$A_s^- \text{ REQ'D } = \frac{299}{4.20(21.5)} = 3.31 \text{ in}^2$$

Try 4 - #9's , A_s^- PRVD = 4.0 in^2

$$L_{dh} = \frac{1200(1.125)}{\sqrt{4000}}\left(\frac{3.31}{4}\right) = 17.7" \approx (20" - 2" \text{ cover} = 18")$$

Use 4 - #9's, Top and Bottom

$$A_s^+ \text{ REQ'D } = \frac{135}{4.20(21.5)} = 1.50 \text{ in}^2 < 4.0 \text{ in}^2 \qquad \text{O.K.}$$

- Check Lateral Bending due to Friction & Anchor Loads

- Friction : Member 5, JT. 4, ; Comb. 2, $V = 12.7^K$
  $V = 0$ @ $x = 12.7/0.8 = 15.88'$, $M_U = 0.1\,(12.7 \times 15.88 - 0.8 \times 15.88^2 / 2) \times 1.4 = 14.1^{\text{k-ft}}$

- Anchor : $M_U = 1.4 \times 2^K \times (30/4 \times 2) = 42.0^{\text{k-ft}}$ ⇐ Governs

- Friction + 30% Anchor : $M_U = 14.1 + 0.3 \times 42.0 = 26.7^{\text{k-ft}}$

$$F = \frac{24(16.5)^2}{12000} = 0.545$$

$$K_U = \frac{42.0}{0.545} = 77.1, \quad a_U = 4.44, \quad A_s \text{ REQ'D} = \frac{42.0}{4.44(16.5)} = 0.58 \text{ in}^2$$

Note: Since max lateral bending moment occurs @ midspan and max vertical bending moment occurs @ JTS. - Assume 1- #9 top and 1 - #9 bottom bar are available at midspan for lateral bending.
A_s PROV'D = 2 x 1.0 = 2.0 in^2 > 0.56 in^2 O.K.

SAMPLE DESIGN 2 : CONCRETE PIPERACK DESIGN

Beam Shear

Design Case : Member 5 @ JT. 3, Comb. : 15, $V_U = 41.03^K$, @ d, $V_U = 33.29^K$

$$\phi V_C = 0.85\,(2)\sqrt{4000}\left(\frac{21.5(20)}{1000}\right) = 46.23^K > 33.29^K$$; Use min. shear reinforcement.

$S_{MAX} = 21.5/2 = 10.75"$, Use 10"

$$A_v \text{ REQ'D} = 50\frac{b_w s}{f_y} = 50\frac{20(10)}{60,000} = 0.167 \text{ in}^2$$

Use #3 Stirrups @ 10" c/c over entire length of beam.

Columns

Design Case : member 2 @ JT.2, Comb. 13, $P_U = 64.53^K$, $V_U = 23.33^K$, $M_U = 236.73^{k\text{-ft}}$

- Check Slenderness

$$\Psi_B = \frac{\Sigma\frac{EI_c}{L_c}}{\Sigma\frac{EI_B \times 0.5}{L_B}} = \frac{\frac{13333}{18} + \frac{13333}{6}}{\frac{11520}{30}} = 7.7, \quad \varphi_n = 1.0 \text{ (Fixed End)}$$

$$K = 1.8, \quad \frac{KL}{r} = \frac{1.8(18)12}{0.3(20)} = 64.8 > 22, \text{ also } 64.8 < 100$$

Therefore, consider slenderness

$$\beta_{db} = \frac{134.46}{134.46} = 1.0$$ (Dead load exceeds loads with wind)

$$EI_b = \frac{E_c I_g / 2.5}{1 + \beta_d} = \frac{(3600\,(13333)/2.5)}{1+1} = 9,600,000 \text{ K-in}^2$$

$$p_{cb} = \frac{\pi^2 EI}{(KL_c)^2} = \frac{\pi^2(9,600,000)}{(1.8\text{x}18\text{x}12)^2} = 627^K$$

$$\delta_b = \frac{C_m}{1 - \frac{p_u}{\phi(p_{cb})}} = \frac{1.0}{1 - \frac{64.53}{0.7(627)}} = 1.17$$

$\beta_{ds} = 0$ (No sustained lateral loads)

$EI_s = 2 \times EI_b = 2 \times 9,600,000 = 19,200,000 \text{ K-in}^2$

$p_{cs} = 2 \times 627 = 1,254^K$

$$\delta_s = \frac{1.0}{1 - \frac{64.53}{0.7(1254)}} = 1.08$$

SAMPLE DESIGN 2 : CONCRETE PIPERACK DESIGN

$m_c = \delta_b m_{b2} + \delta_s m_{2s} = 1.17(10.89) + 1.08(236.73 - 10.89(0.9/1.4)) = 12.7 + 248.1 = 260.8$

$\gamma = \dfrac{20-4}{20} = 0.80$

$\dfrac{p_u}{A_g} = \dfrac{64.5}{(20)^2} = 0.16\,, \quad \dfrac{m_c}{h\,A_g} = \dfrac{260.8(12)}{(20)^3} = 0.39$

From Interaction Charts : ρ REQ'D = 0.020

A_s REQ'D = $0.020 \times 20^2 = 8.0\ in^2$

A_s PROV'D = $12.0\ in^2$

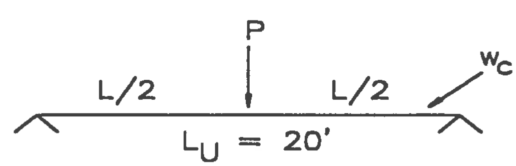
Use 8 - #9's with #3 ties @ 12" c/c

Air Cooler Support Beam Design

Note : A more detailed analysis is necessary where equipment support locations & loads are available, especially the location of beam splice points and the vertical deflection of beams.

P = 50 (DL) + 5 (LL) + 4.5 (WL) = 59.5^K , w_b = 0.50 KLF

M = $59.5(20)/4 + 0.5(20)^2/8 = 323^{\text{k-ft}}$



Use W24 x 104 Beam (M_{Allow} = 435 $^{\text{k-ft}}$)

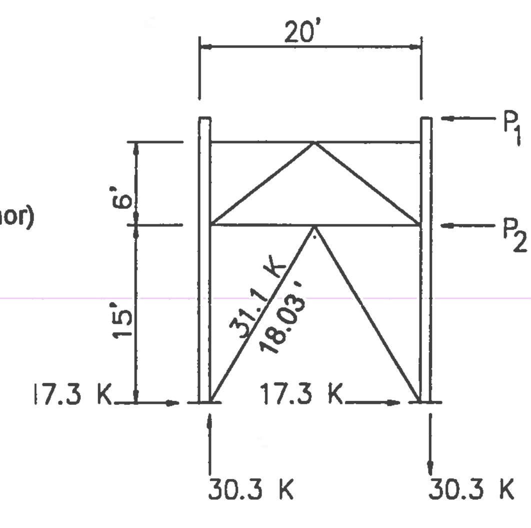
Braced Bay
Longitudinal Force
Note : Wind calculations not shown for brevity. See technical practice 670.215.1215 for procedures.

P_1 = 18 + 4.4 = 22.4^K , P_2 = 10.6^K , Σ = 33^K

Friction
(Use 10% of beam end shear for piping oper. load+30% anchor)

$P_2 = (19.23 \times 0.1 + 0.3 \times (3 \times 2)/2) \times 7$ bays = 19.8^K
$P_1 = (12 \times 0.1 + 0.3 \times (3 \times 2)/2) \times 7$ bays = 14.7^K

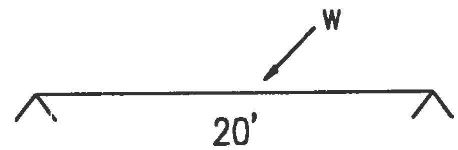
$\Sigma = 19.8 + 14.7 = 34.5^K$ $\Leftarrow$ Governs



SAMPLE DESIGN 2 : CONCRETE PIPERACK DESIGN

Strut (Beam wt = 0.03 KLF)

$$w = \frac{19.23}{\left(\frac{30}{2}\right)} \times 0.5 + 0.03 = 0.67 \text{ KLF}$$



$M = 0.67\,(20)^2 / 8 = 33.5\ ^{'K}$, $P = 22.4^K$

$$\Sigma \frac{fi}{Fi} = 22.4/95 + 33.5/52 = 0.88 \leq 1.0 \text{ O.K.}$$

Use W10x33 Beam, (P_{Allow} = 95^K , M_{Allow} = $52^{k\text{-}ft}$

Bracing

P_{Max} = 31.10^K , $(KL)_Y = (KL)_X$ = 18'

Use 2L - 5x3-1/2x5/16 (LLV) (P_{Allow} = 32^K)

Connections

Air Cooler Support Beam (AISC, ASD Part 4, Connectons)

- Splice : V_{Max} = 64.5^K , # Bolts REQ'D = 64.5/9.3 = 7 (Bolts are in single shear)

For 1/2" PL , L REQ'D = $\frac{64.5}{(0.5)14.4}$ = 9.0" , L PROV'D = 20-1/2" (For 7 Bolts)

Weld : D REQ'D = $\frac{64.5}{20.5(0.928)}$ = 4/16 th's , Use 5/16"

Use PL 1/2" x 6" x 20-1/2" with 7 - 3/4" A325N bolts on one beam & 5/16" Fillet weld on other beam.

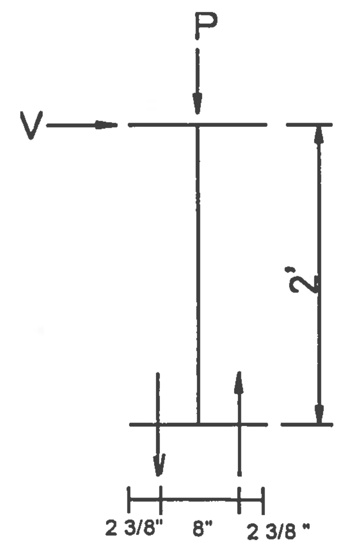
Provide Flange PL to develop 50% of the flange capacity.

Bolts REQ'D = $\frac{12.75(0.75)24}{2(9.8)}$ = 12 bolts

Use PL 3/4" x 13" x 21" with 24 bolts., Typ. top and bottom

SAMPLE DESIGN 2 : CONCRETE PIPERACK DESIGN

Beam Seat Connection



$P = 0.5 \times 20 + 50 - 4.5 = 55.5^K$ min
$= 0.5 \times 20 + 50 + 5 + 4.5 = 69.5^K$ max

$V = 6 + 3.4 = 9.4^K$ (3.4^K wind on beam)

- Required bearing length for web yielding :
N = (69.5 - 44.6) / 11.9 = 2.1" - small

- Required bearing length for web crippling:
N = (69.5 -62.5) / 4.24 = 2.1" - small

Beam seat O.K. in all cases.

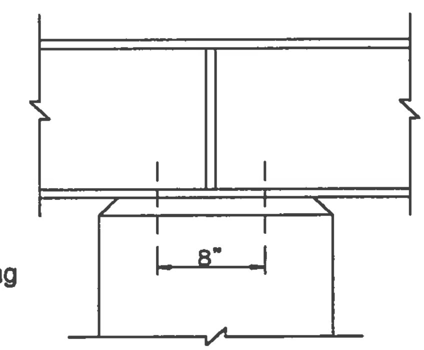
Anchor Bolt Design

$$T_{AC} = \frac{9.4(2)12}{2(8)} - \frac{55.5}{4} = 0.2^K$$

Note : Anchor bolt design not shown for brevity. See technical practice 670.215.1207 for procedures.

Use 4 - 1" ϕ Anchor Bolts

Provide 1/2" Stiff PL @ Centerline of Bent to control shear stresses in web.



Base Plate Design

Note : Base plate design will not be shown for brevity. See technical practice 670.215.1208 for procedures.

Note : The base plate must be mechanically connected to the reinforcing steel in the concrete column. The most likely method is to weld rebar to the base plate as shown in PCI Design Handbook; however an alternate detail is shown here to avoid welding rebar.

Use PL 1-1/2" x 28" x 28" with 6 - 1-1/2" ϕ anchor bolts.

Steel Beam / Bracing to Concrete Column Connections

Note : For brevity, the design of these connections is omitted. The details shown were designed for the compression and/or tension force component perpendicular to the face of the column in combination with the shear force parallel to the face of the column, using headed studs. For these types of connections, headed studs, rebar, embedded inserts, anchor bolts, or sleeved threaded rods can be used. For design procedures, see PCI Design Handbook. Also manufactures publications provide design procedures, capacities, and constraints.

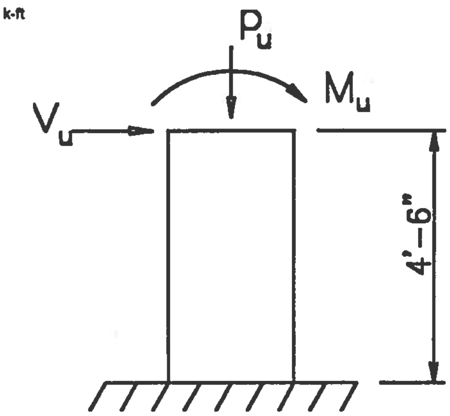
SAMPLE DESIGN 2 : CONCRETE PIPERACK DESIGN

Foundation Design

(f'_c = 3 KSI)

Pier Design (Use 3' - 0" Square Pier , Say height is 4' - 6")

@ Joint 2, load comb. 14, $P_u = 135.49^K$, $V_u = 25.49^K$, $M_u = 253.14^{k\text{-}ft}$



$$F = \frac{36(33.5)^2}{12000} = 3.37$$

$$K_u = \frac{253.14}{3.37} = 75, \quad a_u = 4.42$$

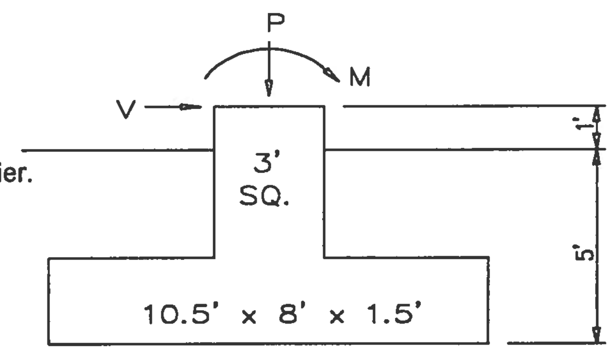
$$A_s \text{ REQ'D} = \frac{253.14}{4.42(33.5)} = 1.71 \text{ in}^2 \text{ / face} \quad (\times 1.33 = 2.27 \text{ in}^2 \text{ / face})$$

A_s MIN = 0.0033 x 36 x 33.5 = 3.98 in^2 / face

A_s MIN COL. = 0.005 x 36^2 = 6.48 in^2 (Total)

> Use 12 - #7's with #3 ties @ 12" c/c
> (A_s PROV'D = 12 x 0.6 = 7.2 in^2 (Total))
> (A_s PROV'D = 4 x 0.6 = 2.4 in^2 / face)

Footing Design



Try 10'-6" x 8'-0" x 1'-6" thick footing

Design Cases : Transverse Forces Applied @ top of pier.

Unfactored : 1. Joint 1 , Load 9 , $P = 131.62^K$, $V = 20.20^K$, $M = 197.80^{k\text{-}ft}$
2. Joint 2 , Load 7 , $P = 79.16^K$, $V = 17.56^K$, $M = 180.20^{k\text{-}ft}$

Pier Weight = 3 x 3 x 4.5 x 0.15 = 6.1^K
Footing Weight = 10.5 x 8 x 1.5 x 0.15 = 18.9^K
Soil Weight = $(10.5 \times 8 - 3^2) \times 3.5 \times 0.12 = 31.5^K$
$\Sigma = 56.5^K$

$P_1 = 131.6 + 56.5 = 188^K$
$M_1 = 197.8 + 20.2 \times 6 = 319^{k\text{-}ft}$

e = 319/188 = 1.70' < B/6 = 10.5'/6 = 1.75'

$$SB_{GROSS} = \frac{188}{84}\left(1 \pm \frac{6(1.70)}{10.5}\right) = 4.41 \text{ KSF} < 4.6 \text{ KSF allowable} = (1.33 \times 3 + 5 \times 0.12) \quad \text{O.K.}$$

$P_2 = 79.16 + 56.5 = 136^K$
$M_2 = 180.2 + 17.56 \times 6 = 286^{k\text{-}ft}$

SAMPLE DESIGN 2 : CONCRETE PIPERACK DESIGN

$e = 286/136 = 2.1' > B/6 = 10.5'/6 = 1.75'$

$$SB_{GROSS} = \frac{136}{84}\left(\frac{4(10.5)}{3(10.5-2(2.1))}\right) = 3.60 \text{ KSF} < 4.6 \text{ KSF} \quad \text{O.K.}$$

Check Stability for Case 2

$$M_{RS} = (\underset{P_{DL}}{96.04} + \underset{\text{fdn wt}}{56.5})\,5 + \underset{V_{DL}}{1.2} \times 6 + \underset{M_{DL}}{7.77} = 778'^{K}$$

$$M_{OT} = \underset{T_{WL}}{16.56} \times 5 + \underset{V_{WL}}{18.58} \times 6 + \underset{M_{WL}}{183.10} = 377'^{K}$$

SR = 778/377 = 2.06 > 1.5 O.K.

Use 10'-6" x 8' -0" x 1'-6" thick footing.

Check Longitudinal Forces

$P_1 = 114.7 + 56.5 + 30.3 = 201.5^K$, $M_{RS} = 684'^K$, $SR = \infty$, $e = 0.52'$, $SB_{Max} = 3.32$ KSF
$P_2 = 108.4 + 56.5 - 30.3 = 134.6^K$, $M_{RS} = 660'^K$, $SR = 2.93$, $e = 0.77$, $SB_{MAX} = 2.53$ KSF
$M_{OT} = 17.3 \times 6 \pm 30.3 \times 4 = 103.8 \pm 121.2 = -17.4^{\text{ k-ft}}, 225^{\text{ k-ft}}$ O.K.

- Footing Reinforcing Steel

Design Cases
- Factored : 1 Joint 1, Load Comb. 15, $P_U = 142^K$, $V_U = 25.5^K$, $M_U = 252^{\text{ k-ft}}$

$P_{U1} = 147 + 1.4 \times 0.75 \times 56.5 = 206^K$
$M_{U1} = 252 + 25.5 \times 6 = 405^{\text{ k-ft}}$
$e = 405/206 = 1.97' > B/6$

$$SB_{GROSS} = \frac{206}{84}\left(\frac{4(10.5)}{3(10.5-2(1.97))}\right) = 5.23 \text{ KSF}$$

$$x = \frac{2(206)}{8(5.23)} = 9.85'$$

$w = (3.5 \times 0.12 + 1.5 \times 0.15) \times 1.4 \times 0.75 = 0.68$ KSF

$$SB_F = \left(\frac{9.85-3.75}{9.85}\right) \times 5.23 = 3.24 \text{ KSF}$$

M_U @ face of pier $= (3.24 - 0.68)(3.75)^2 / 2 + 1.99(3.75)^2 /3 = 27.3^{\text{ k-ft}} / \text{ft}$

$d = 18 - 3.5 = 14.5"$

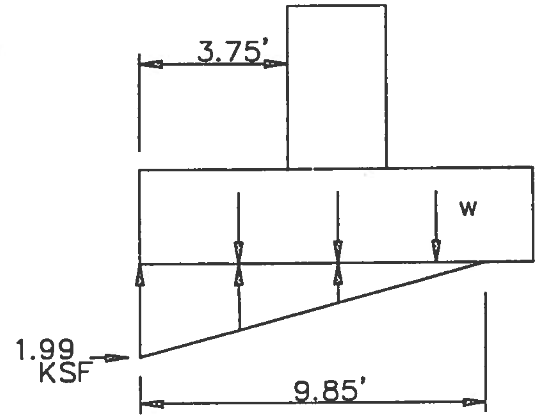
$$F = \frac{12(14.5)^2}{12000} = 0.210$$

$K_U = 27.3/0.21 = 130$, $a_U = 4.37$

FLUOR DANIEL

SAMPLE DESIGN 2 : CONCRETE PIPERACK DESIGN

A_s REQ'D = $\frac{27.3}{4.37(14.5)}$ = 0.43 in² / ft (x 1.33 = 0.57 in² / ft)



Use #7's @ 12" c/c EW - T & B

Note: Top steel required for negative bending due to uplift.

- Check Footing Thickness

(7/8 in the following equation is from the #7 Bars in the pier)

L_{dh} = (1200 x (7/8) $\sqrt{3000}$) x 0.7 x 2.27/2.4 = 12.69"

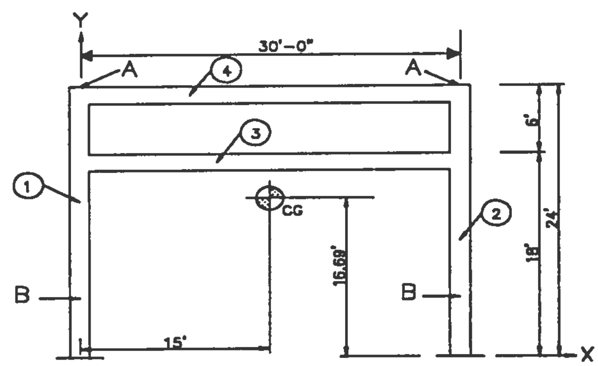
Footing thicknes req'd = 12.69" + 2 x (7/8) + 3 = 17.44" < 18" O.K.

V_u @ face of pier = (3.24 - 0.68)3.75 + 1.99 x 3.75/2 = 13.3^K

ϕV_c = 0.85 x 2 x $\sqrt{3000}$ x 12 x 14.5 = 16.2^K > 13.3^K O.K.

Punching shear is O.K. by inspection

Use 1'-6" Footing Thickness



Lifting Arrangements / Lugs

Wt. of concrete bent
Wt. Cols. = 1.67² x 24 x 0.15 = 10.04^K / each
Wt. Bms. = 1.67 x 2 x 28.33 x 0.15 = 14.19^K / each

Total weight = 2 (10.04 + 14.19) = 48.46^K

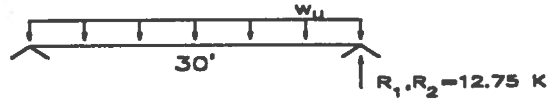
Determine C.G.

Item	WT	X	Y	WT*X	WT*Y
1	10.04	0	12	0	120.48
2	10.04	30	12	301.2	120.48
3	14.19	15	17	212.85	241.23
4	14.19	15	23	212.85	326.37
Σ	48.46			726.9	808.56

SAMPLE DESIGN 2 : CONCRETE PIPERACK DESIGN

$\bar{X} = \frac{726.90}{48.46} = 15.00'$ $\bar{Y} = \frac{808.56}{48.46} = 16.69'$

Locate Lifting Points



Use air cooler support beam seat connection for lifting points "A".

Use y = 24 - (24 - 16.69) x 2 = 9.38' , say 9' 4-1/2" for lifting points "B".

<u>Check Reinforcement for Bending in Horizontal Bent</u>

- Beam : $w_U = 1.67 \times 2 \times 0.15 \times 1.7 = 0.85$ KLF

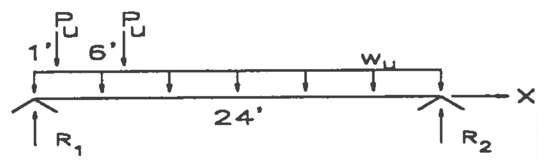
$M_U = 0.85 \times 30^2 / 8 = 96^{k\text{-}ft}$

$$A_S \text{ REQ'D} = \frac{96(12)}{(0.9)^2(60)16.5} = 1.44 \text{ in}^2 < 2 - \#9\text{'s } (A_S = 2.0 \text{ in}^2) \quad \text{O.K.}$$

- Column : $P_U = 12.75^K$

$w_U = (1.67)^2 \times 0.15 \times 1.7 = 0.71$ KLF

$M_{U2} = 0.71(9.38)^2 / 2 = 31.23^{k\text{-}ft}$



$$R_1 = 12.75 \times \frac{(13.62 + 7.62)}{14.62} + 0.71 \times 14.62/2 - 31.23/14.62 = 21.58^K$$

$R_2 = 2 \times 12.75 + 0.71 \times 24 - 21.58 = 20.96^K$

V = 0 @ x = 7'-0",

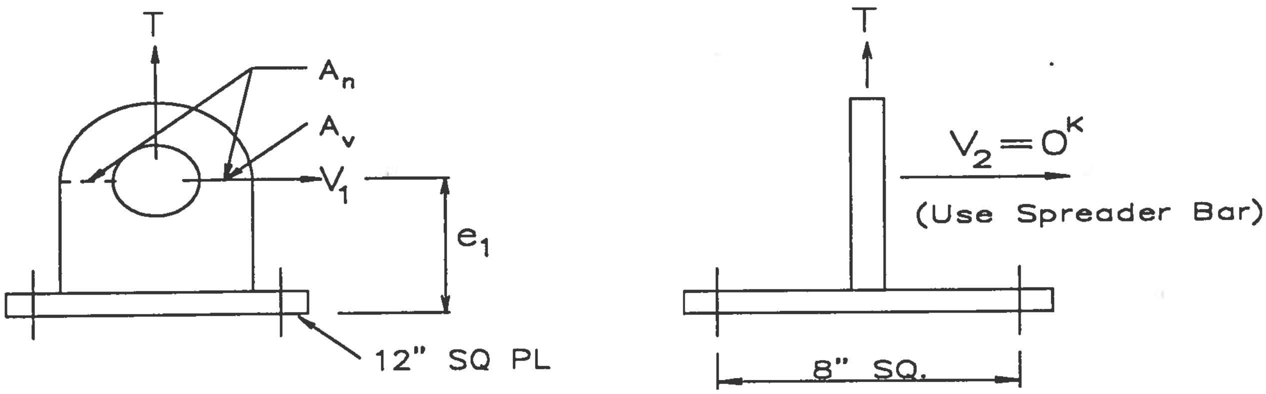
$M_{U\,Max} = 21.58 \times 7 - 12.75 \times 6 - 0.71 \times (7)^2 / 2 = 57.2^{k\text{-}ft}$

$$A_S \text{ REQ'D} = \frac{57.2(12)}{(0.9)^2(60)17.5} = 0.81 \text{ in}^2 < 3 - \#9\text{'s } (A_S = 3.00 \text{ in}^2) \text{ O.K.}$$

<u>Connections @ lifting PT. "A"</u>

$T = 48.46/2 \times (1 + 25\% \text{ impact}) = 30.3^K$ Use 25% impact per FD construction $V_1 = 30.3 / 2 = 15.2^K$

SAMPLE DESIGN 2 : CONCRETE PIPERACK DESIGN



Check Anchor Bolts (4 - 1"ϕ A307 bolts)

$P_c = \phi 4 \sqrt{f'_c}\, A_C = 0.65(4)\sqrt{4000}\,\frac{(20)^2}{1000} = 65.8^K$ ⇐ Governs

P_T = #bolts x A_b x F_Y = 4 x (0.606) x (36) = 87.3^K

P_c / T = 65.8 / 30.3 = 2.2 > 1.7 O.K. for direct tension

$V_c = 65.8 \times 0.55 = 36.2^K$

$V_U = 1.7 \times 15.2 = 25.8^K < V_c$ O.K. for shear only

$T_U = (36.2 - 25.8) / 0.55 = 18.9^K$

$e_{1\ allow} = (T_U / V_U) \times 8 = (18.9 / 25.8) \times 8 = 5.9"$

Use e_1 = 5 - 1/2"

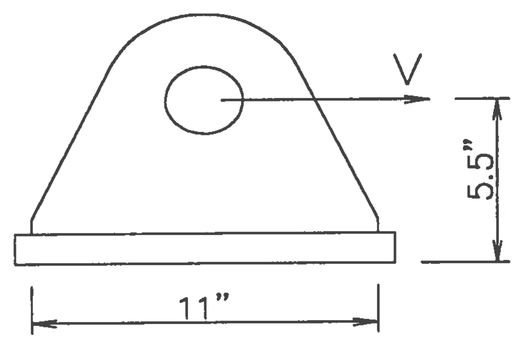
Base Plate

$M_{PL} = (30.3 \times 8) / 4 = 60.6^{\ k\text{-}ft}$

t_{PL} REQ'D = $\sqrt{\frac{6(60.6)}{12(27)}} = 1.06"$

SAMPLE DESIGN 2 : CONCRETE PIPERACK DESIGN

Use 1-1/4" Base Plate



SAMPLE DESIGN 2 : CONCRETE PIPERACK DESIGN

Lug Plate and Pin

$\boxed{\text{Use 1-1/4" Lug Plate}}$

A_N REQ'D = $\frac{30.3}{16.2}$ = 1.87 in^2

A_V REQ'D = $\frac{15.2}{14.4}$ = 1.06 in^2

Note : Tension is carried by plate on both sides of pin, shear is carried by one side only.

Total A_V REQ'D = 2 x 1.06 = 2.12 in^2 ⇐ Governs

Pin Diameter Required, D = $\sqrt{\frac{30.3}{2(10)} \times \frac{4}{\pi}}$ = 1.57"

Note : Pin is in double shear, Allowable shear stress for A307 material in bearing conn. is 10 KSI

$\boxed{\text{Use 2"}\phi\text{ pin with min. A307 material.}}$

$F_P = 0.9\ F_Y = 0.9 \times 36 = 32.4$ KSI

Bearing Area Required = Dt = 30.3/32.4 = .94 in^2

Dt PROV'D = 2 x 1.25 = 2.5 in^2 , No hole reinforcement required.

Lug Plate width at centerline of hole = 2.125 + 2.12/1.25 = 3.82"

$\boxed{\text{Use 3" radius to out side of lug plate. (width = 6", O.K.)}}$

Lug plate to Base plate weld

- For Tension

D REQ'D = $\frac{30.3}{2(11)0.928}$ = 2/16 ths

- For Shear (eccentric load on weld)

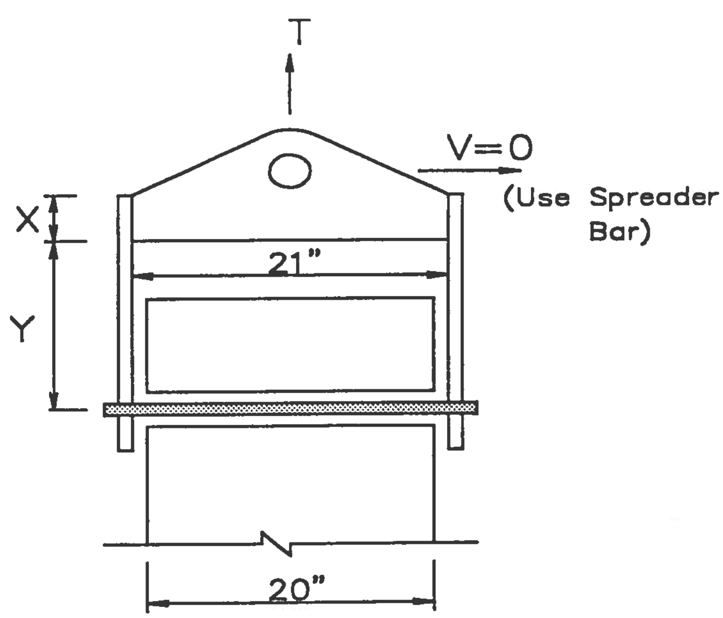
p = V = 15.2^K , L =11", K =0

a = (5.5 - 1.25) / 11 =0.4, C_1 =1.0

From table XIX , p4-75, AISC C = 0.939

D REQ'D = $\frac{15.2}{0.939(1.0)(11)}$ = 2/16 ths

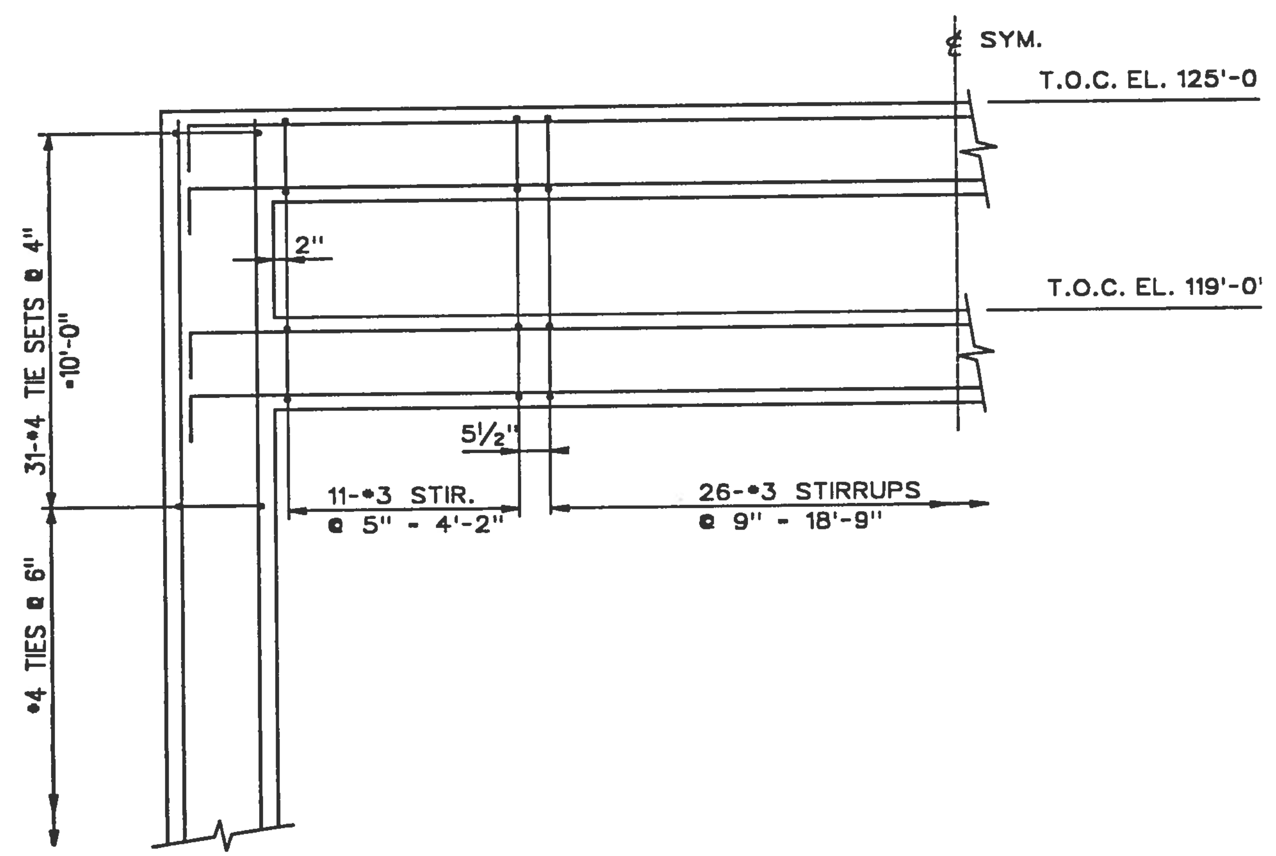
$\boxed{\text{Use 5/16" weld (min)}}$



SAMPLE DESIGN 3 : CONCRETE PIPERACK DESIGN WITH SEISMIC DESIGN

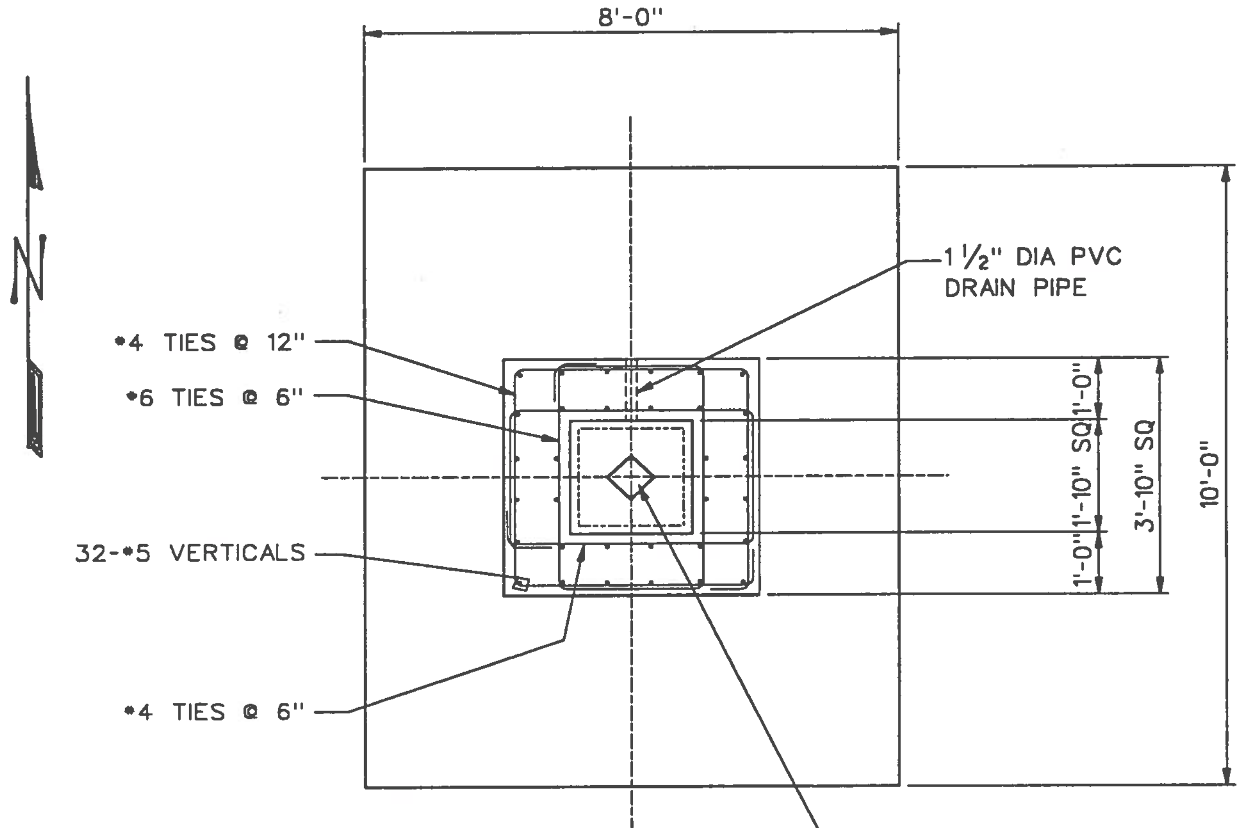
GIVEN

- Refer to SK-1 through SK -7 of sample design #2. The plan, elevation, sections, & details shown in sample design #2 are acceptable for this design except as shown below. Note that the base plate detail in sample design #2 is no longer required since a socket is used for this design.

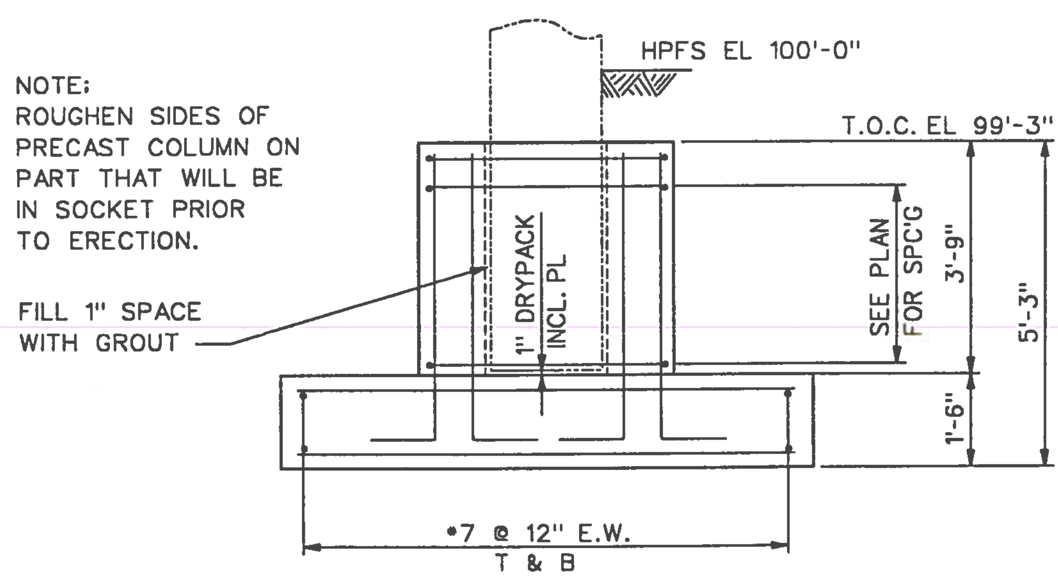


PARTIAL ELEVATION/SECTION @ TYPICAL BENT

SAMPLE DESIGN 3 : CONCRETE PIPERACK DESIGN WITH SEISMIC DESIGN



PLAN



SECTION

SAMPLE DESIGN 3 : CONCRETE PIPERACK DESIGN WITH SEISMIC DESIGN

Socket Grouting Notes

1. After erection of concrete bents , including plumbing, plug drain pipe and fill socket with water. Allow water to remain for 3 hours.

2. Unplug drain pipe , pump and blow out the water just prior to grouting. Leave concrete surfaces moist but not wet. Replug drain pipe.

3. Install temporary grout dams (made of styrofoam or similar material) on opposite faces of a column and pour grout (sand cement or non-shrink - per job spec) from side only until grout starts flowing out on the other side, indicating complete filling of the underside of the column.

4. Remove dams and continue filling spaces with grout.

Design Data

This sample design demonstrates concrete design using ACI - 318 - 89, Ch. 21 in a UBC seismic zone 4. It also includes the design of a socket foundation.

References and materials from sample design #2 are used for this design.

Design Loads

- Use gravity loads from sample design #2.
- Use wind loads from sample design #2 as earthquake loads. For calculations of actual earthquake loads, see technical practice 670.215.1216 for procedures.
- Load combinations are the same as sample design #2.

Design Model

Same as sample design #2.

REQUIRED

Detail/ Design concrete members and connections used in sample design #2 for seismic forces.

SOLUTION

Member Design

Beams

The design performed in sample design #2 is valid except for the following requirements.

- $P_{U\,max} = 37.0^K \leq A_g f_c /10 = (20 \times 18 \times 4)/10 = 144^K$ O.K.

SAMPLE DESIGN 3 : CONCRETE PIPERACK DESIGN WITH SEISMIC DESIGN

- Clear span = 30 - 1.67 = 28.33" ≥ 4d = 4 x (15.5/12) = 5.77' O.K.
- b/h = 20/18 = 1.11 ≥ 0.3 O.K.
- b = 20" ≥ 10" & ≤ b_{col} + 2 x 3/4 x h_{bm} = 20 + 2 x 0.75 x 18 = 47" O.K.

- Longitudinal Reinforcement (Code minimum requires 2 bars continuous at top and bottom)

- $A^-_s = 4.00 \text{ in}^2$ & $A^+_s = 4.00 \text{ in}^2 \geq (200 b_w d / f_y) = \dfrac{200(20)16}{60000} = 1.07 \text{ in}^2$ O.K.
- $\rho^- = 0.0125$ & $\rho^+ = 0.0125 \leq 0.025$ O.K.
- $A^+_s = 4.00 \text{ in}^2 > A^-_s / 2 = 2.00 \text{ in}^2$ O.K.

Use 4 - #9's - Top & Bottom (A_s PROV'D = 4.0 in²)

$L_d = 0.04 \times 1.00 \times \dfrac{60000}{\sqrt{4000}} \dfrac{2}{4} = 19" > 18"$ PROV'D , therefore hook bottom bars.

- Transverse Reinforcement

- Place stirrups starting @ 2" from col. face with 5" spacing over 2 x h = 2 x 24 = 48" , otherwise use 9" spacing.

- Shear Strength

$A_g f_c' / 20 = (20 \times 24 \times 4) / 20 = 96^K > P_{U\,max} = 37^K$, therefore $V_C = 0$

$$a = \frac{A_S(1.25)f_Y}{0.85 f_c' b} = \frac{4.0(1.25)60}{0.85(4)20} = 4.41"$$

$M_{pr1} = M_{prz} = A_S \times 1.25 f_Y (d - a/2) = 4 \times 1.25 \times 60 \times (21.5 - 4.4/2) / 12 = 483'^K$

$$V_E = \frac{M_{pr1} + M_{prz}}{L} \pm V_{U\,gravity} = \frac{2(483)}{28.33} \pm 19.24 = 53.3^K$$

$$A_v \text{ REQ'D} = \frac{V_e s}{f_Y d} = \frac{53.3(4)}{60(21.5)} = 0.17 \text{ in}^2$$

A_v PROV'D = 2 x 0.11 = 0.22 in² (#3 stirrup)

Use #3 stirrups placed as described above.

Columns

The design made in sample design #2 is valid except for the following requirements.

- $P_{U\,max} = 160.58^K = A_g f'_c /10 = (20 \times 20 \times 4)/10 = 160^K$ Flexural strength is O.K.
  (160.58^K from member 1, JT.1, Load comb. 11)
- $\rho = 0.02 > 0.01$ & < 0.06 O.K.

SAMPLE DESIGN 3 : CONCRETE PIPERACK DESIGN WITH SEISMIC DESIGN

- $A_{Sh} = 0.3 (s\, h_c\, f_c' / f_{gh}) \times [(A_g / A_{ch}) - 1]$
  $= 0.3 (4 \times 17.38 \times 4/60) \times [(400/(17)^2 - 1] = 0.53 \text{ in}^2$ per ACI

or $A_{Sh} = 0.09\, s\, h_c\, f_c' / f_{yh} = 0.09 \times 4 \times 17.38 \times 4/60 = 0.42 \text{ in}^2$ per ACI

or $A_{3h} = 0.12\, s\, h_c\, f_c' / f_{yh} = 0.12 / 0.09 \times 0.42 = 0.56 \text{ in}^2$ per UBC ⇐ Governs

For 3 bars , A'_{sh} REQ'D = $0.56 \text{ in}^2 / 3 = 0.19 \text{ in}^2$ / bar

$L_o = 20"$ ⇐ Governs at Top of column

or $L_o = 16.5(12) / 6 = 33"$ ⇐ Governs at Bottom of column

or $L_o = 18"$

Use #4 ties with #4 cross ties @ 4" c/c within L_o region. Use 6" c/c spacing outside of L_o region.

Beam / Column Joints

- Beam Longitudinal Reinforcement

Top and Bottom Steel : 4 -#9's

$$L_{dh} = \frac{f_Y d_b}{65\sqrt{f_c'}} = \frac{60000(9/8)}{65\sqrt{4000}} = 16.4" < 18.5" \qquad \text{O.K.}$$

- Transverse Reinforcement

Use #4 ties with #4 cross ties @ 4" spacing on column within joint.

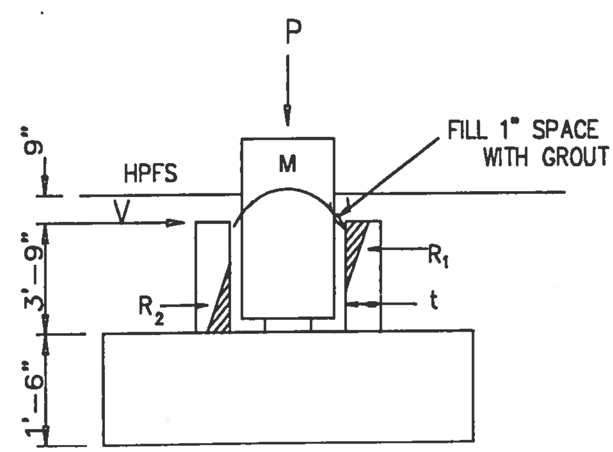
- Shear Strength = $12\sqrt{f_c'}\, A_U = 12\sqrt{4000}\, \frac{20^2}{1000} = 304^K$ >> Actual and factored shear across the joint. O.K.

<u>Socket</u> <u>Design</u>

Design Data

Column Size	: 20" x 20"
Column Reinforcement	: 12 - #9 verts.
f_c'	: 3.0 KSI
f_Y	: 60.00 KSI

Grout type : Non-shrink cement based grout between the socket and the column.



FLUOR DANIEL

SAMPLE DESIGN 3 : CONCRETE PIPERACK DESIGN WITH SEISMIC DESIGN

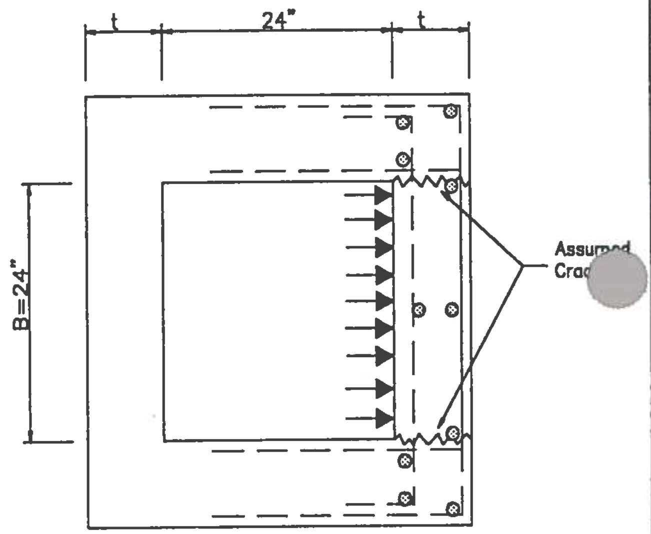
Design Assumptions

Assume that column shear, V, and moment ,M, are resisted by the couple , R, acting on the forward and leeward socket walls.

Axial load , P, transfers directly to the top of the footing. Therefore, socket walls carry no axial load.

Zero adhesion is assumed between the column and the socket walls, since cement based grout may be used.

Couple, R, is transfered to the end walls through horizontal bending of forward and leeward walls and through shear friction across assumed cracks.



Socket Dimensions

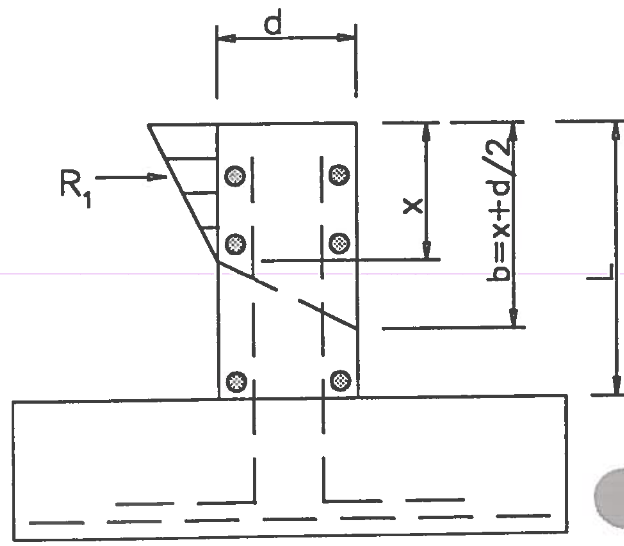
Socket wall thickness (t) =	16.00 in.

Provide depth of socket for development of #9 column reinforcing.

For a #9 in tension, top bar, f_c' = 4.0 KSI

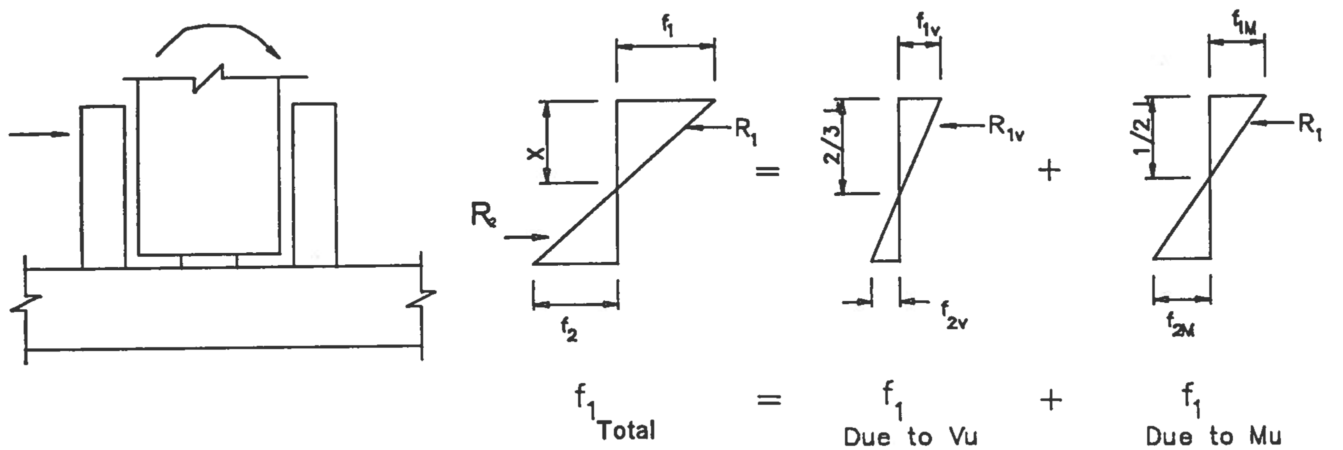
l_d =	42.00 in.
$\frac{A_S REQ'D}{A_S PROV'D} = \frac{12}{12(1.0)}$ =	1.0
l_d REQ'D =	42.00 in

Hpp el. @ CL of P/R =	100.00 ft
Less: X-slope to CL of col. =	0.17 ft
T.O. Pvmt el. @ CL of col. =	99.83 ft
Less : Slab Thickness =	0.5 ft.
B.O. Slab el. =	99.33 ft
Less : Expansion joint material =	0.08 ft
T.O. Socket el. =	99.25 ft
Less : l_d REQ'D =	3.50 ft
B.O. concrete reinforcement el. =	95.75 ft
Less : Concrete cover =	0.17 ft
B.O. concrete col el. =	95.58 ft
Less : Grout and level PL thickness =	0.08 ft
Top of footing el. =	95.50 ft
Socket depth (L) =	3.75 ft



SAMPLE DESIGN 3 : CONCRETE PIPERACK DESIGN WITH SEISMIC DESIGN

Design Model



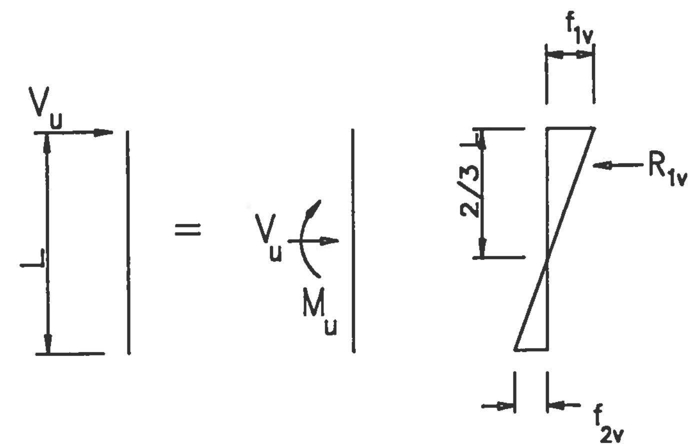
Determine f_1 due to V_U :

Transfer V_U @ top of socket to CL of socket :

$$M = \frac{V_U L}{2}$$

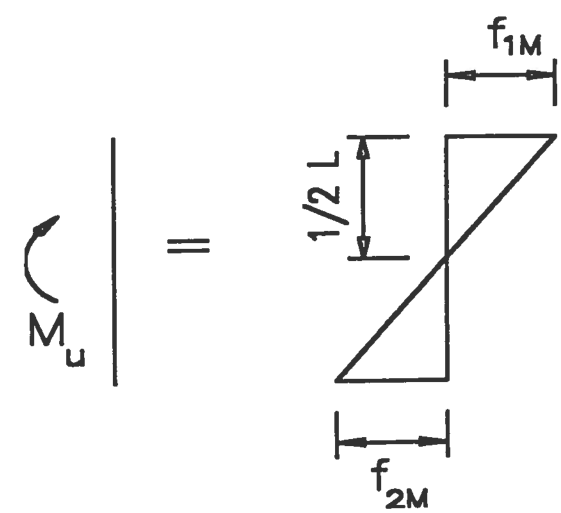
$$f_{1V} = \frac{V_U}{BL} + \frac{6M}{BL^2} = \frac{V_U}{BL} + \frac{6V_U L}{2BL^2} = \frac{4V_U}{BL}$$

$$f_{2Y} = \frac{V_U}{BL} - \frac{3V_U}{BL} = \frac{-2V_U}{BL}$$



Determine f_1 due to M_U :

$$f_{1M} = f_{2M} = \frac{6M_U}{BL^2}$$



FLUOR DANIEL

SAMPLE DESIGN 3 : CONCRETE PIPERACK DESIGN WITH SEISMIC DESIGN

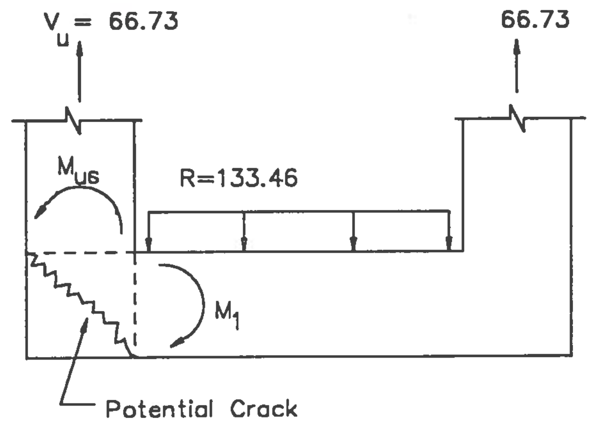
Column Reactions from STAAD III Analysis

Critical transverse load comb. : File : CONCPR, Joint 2, Load 14

M_U @ top of socket =	253.14 ft-kips
V_U @ top of socket =	25.49 kips

Critical longitudinal load comb. :14 , Friction

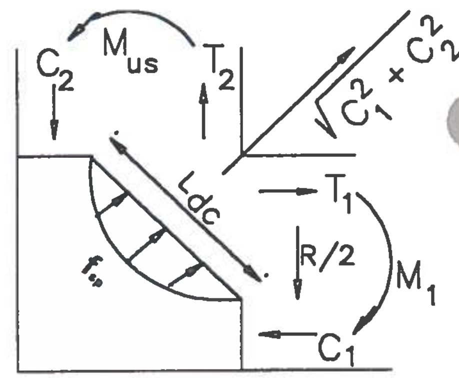
M_U @ top of socket =	0.0 ft-kips
V_U @ top of socket =	24.22 kips



<u>Socket Design - Transverse Direction</u>

Determine bearing pressure distribution on socket wall.

B = Col diam. + 2" grout =	22.00 in
$f_{1V} = 4V_U / BL$ =	102.99 psi
$f_{2V} = -2V_U / BL$ =	-51.49 psi
$f_{1M} = -f_{2M} = 6M_U / (BL)^2$ =	409.12 psi
$f_1 = f_{1V} + f_{1M}$ =	512.11 psi
$f_2 = f_{2V} + f_{2M}$ =	-460.61 psi
ϕ (ACI 9.3.2.4) =	0.70
$f_{P\ allow} = 0.85\ \phi\ f_c'$ =	1785.00 psi O.K.
$x = f_1 / (f_1 - f_2)L$ =	23.69 in
$R_1 = 0.5\ f_1 (B)x$ =	133.46 kips



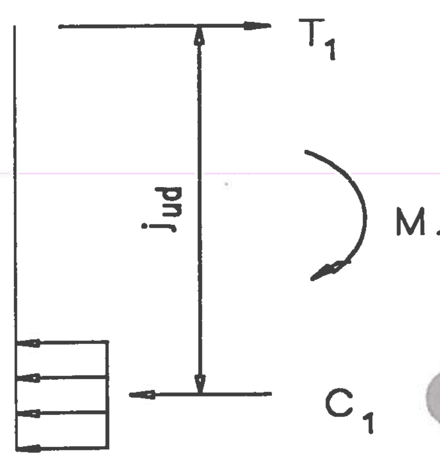
FREE BODY OF CORNER

Check shear stress @ assumed crack section (shear friction)

$V_U = R_1 / 2$ =	66.73 kips
ϕ = (ACI 9.3.2.3) =	0.85
d = (socket wall thk - clr - $0.5d_b$) =	13.5 in
$b_{eff} = x + d/2$ =	30.44 in
$v_U = V_U / (\phi\ b_{eff}\ d)$ =	191.03 psi
$v_{U\ allow} = 0.20\ f_c'$ =	600.00 psi O.K.

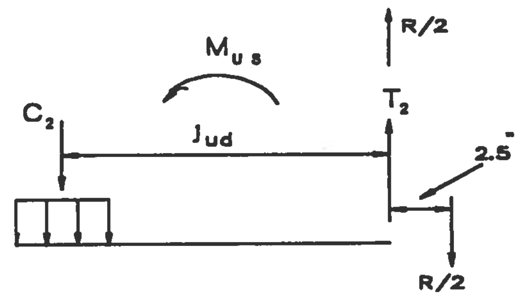
Check wall thickness for diagonal tension cracking @ corner
(Ref : ASCE Structural Journal, June 76, pp 1229 - 1254)

$f_{SP} = 6\sqrt{f'_C}$ =	328.63 psi
$b = b_{eff}$ =	30.44 in
$l_{dc} = 1.1d$ =	14.85 in
$C_{allow} = (2/3)f_{sp}\ b\ l_{dc}$ =	99.04 kips
$R = R_1$ =	133.46 kips
$l = B$ =	22.00 in
$M_1 = R * l / 12$ =	20.39 ft-kips
$F = bd^2 /12000$ =	0.462
$K = M_{US} / F$ =	44.10
j_U = (ACI Handbook Flexure 1.1) =	0.988
$C_1 = T_1 = M_1 / (j_U * d)$ =	18.34 kips



SAMPLE DESIGN 3 : CONCRETE PIPERACK DESIGN WITH SEISMIC DESIGN

$M_{US} = M_1 + (R/2) * 2.5 / 12 =$	34.29 ft-kips
$K = M_1 / F =$	74.17
j_U = (ACI Handbook, Flexure 1.1) =	0.983
$C_2 = T_2 = M_{US} / (j_U * d) =$	31.01 kips
$C = \sqrt{C_1^2 + C_2^2} =$	36.03 kips O.K.



Design Ties

"F" Ties - Design for combined bending (M_{US}) + Tension at joint.

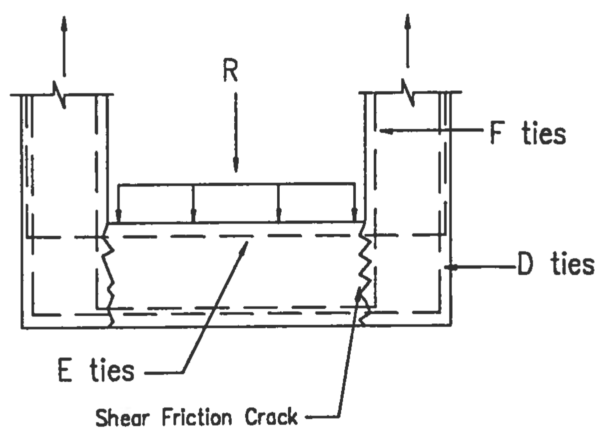
$T = T_2 + R/2 =$	97.74 kips
ϕ = (ACI 9.3.2.2) =	0.90
$A_S = T / (\phi f_Y) =$	1.81 in^2
$A_S / FT = A_S / b =$	0.71 in^2 / ft

$\boxed{\text{Use \#6 ties @ 6" c/c}}$

A_S PROV'D =	0.88 in^2 / ft
l_{dh} = (ACI)	11.50 in
l_{dh} PROV'D =	14.00 in O.K.

"E" Ties - Design for bending (M_1) and for shear friction.

ϕ = (ACI 9.3.2.2) =	0.90
$A_{S1} = T_1 / (\phi f_Y) =$	0.34 in^2
$A_{S1} / FT = A_S / b =$	0.13 in^2 / ft
ϕ = (ACI 9.3.2.3) =	0.85
M_U = (ACI 11.7.4.3) =	1.4
$A_{Vf} = V_U / (\phi f_Y M_U) =$	0.36 in^2
$A_{Vf} / FT = A_{Vf} / b =$	0.14 in^2 / ft
$0.5(A_{S1} / FT + A_{Vf} / FT) =$	0.14 in^2 / ft



$\boxed{\text{Use \#4 ties @ 6" c/c}}$

A_S PROV'D =	0.40 in^2 /ft

"D" Ties - Design for bending at midspan & for shear friction.

$M_U = R * l / 10 =$	24.47 ft-kips
$K = M_U / F =$	52.92
a_U =	4.45
$A_S = M_U / (a_U * d) =$	0.41 in^2
$A_S / FT = A_S / b =$	0.16 in^2 / ft
$\frac{A_{Vf}}{FT} = A_{Vf} - ((A_S \text{ PROV'D "E"}) - \frac{A_{S1}}{FT} =$	0.09 in^2 /ft

$\boxed{\text{Use \#4 ties @ 12" c/c}}$

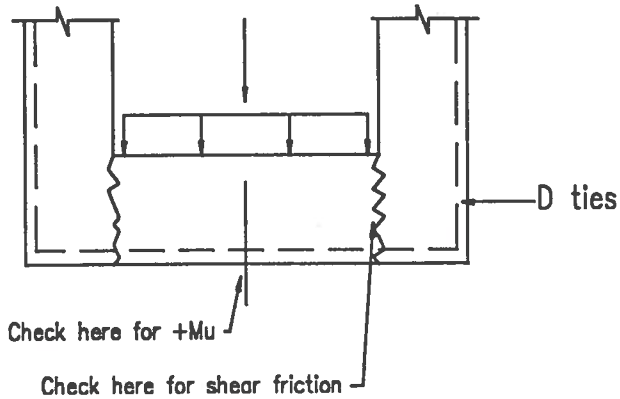
A_S PROV'D =	0.20 in^2 /ft O.K.

SAMPLE DESIGN 3 : CONCRETE PIPERACK DESIGN WITH SEISMIC DESIGN

Design Socket End Walls

$V_U = R_1 / 2$	66.73 kips
l_W = 2 * Socket wall thickness + B =	54.00 in
ϕ = (ACI 9.3.2.3) =	0.85
$d = 0.8 * l_W$ =	43.20 in
$v_U = V_U / (\phi * h * d)$ =	113.58 psi
$v_C = 2\sqrt{(f'_C)}$ =	109.54 psi

Shear reinforcing is required



Horizontal Shear Reinforcing Req'd

P_h = Max of $(v_U - v_C) / f_y$ or 0.0025 =	0.0025	
Max Spc'g = Min of (l_W /5),3h, or 18" =	10.80 in	
Use spacing "s" =	6.00 in	

Use #6 "F" tie @ 6" spacing

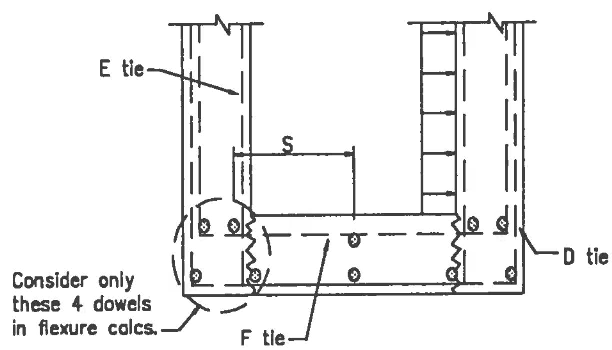
A_V REQ'D = $P_h * s * h$ =	0.24 in^2	
A_V PROV'D = (#6 tie at 6") =	0.44 in^2	O.K.

Vertical Shear Reinforcing Req'd

h_W = h = 16.00 in

$$P_n = \text{Max of } .0025 + \frac{2.5 - h_W}{2(l_W)}(P_h - .0025)$$

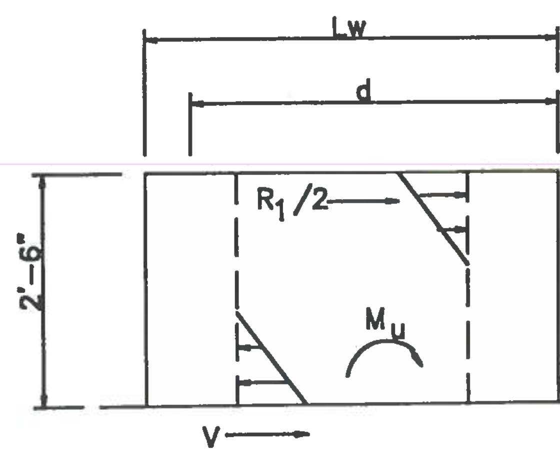
or 0.0025 =	0.0025	
Max Spc'g = Min of (l_W /3), 3h or 18" =	18.00 in	
Use "s" = Dowel Spacing =	9.55 in	
A_V REQ'D = $P_h * s * h$ =	0.38 in^2	
A_V PROV'D = (2 - #6 at 9.55") =	0.88 in^2	O.K.



Check Wall for Flexure

M_U = 0.5(M_U @ T.O. socket + V_U @ T.O. socket * L)

=	174.36 ft-kips	
b = h =	16.00 in	
$d = l_W - h/2$ =	46.00 in	
$F = bd^2 / 12000$ =	2.821	
$K = M_U / F$ =	61.80	
a_U =	4.44	
$A_S = M_U / (a_U * d)$ =	0.85 in^2	
A_S PROV'D = (4 - #6's @ "s") =	1.76 in^2	O.K.



Check Shear Transfer @ T.O. Footing

V_U = 0.5 * V_U @ T.O. socket =	12.745 kips
ϕ = (ACI 9.3.2.3) =	0.85
μ = (ACI 11.7.4.3) =	1.00
$A_{vf} = V_U / (\phi * f_Y * \mu)$ =	0.25 in^2

Use 6 - #5 Dowels each face.

A_{vf} PROV'D = (6 - #6) = 2.64 in^2

SAMPLE DESIGN 3 : CONCRETE PIPERACK DESIGN WITH SEISMIC DESIGN

Socket Design - Longitudinal Direction

Determine Bearing Pressure Distribution on Socket Wall

B = (Col Dim.) + (2" Grout) =	22.00 in
$f_{1v} = 4V_U / (BL)$ =	97.86 psi
$f_{2v} = -2V_U / (BL)$ =	-48.93 psi
$f_{1M} = -f_{2M} = 6M_U / (BL^2)$ =	0.00 psi
$f_1 = f_{1V} + f_{1M}$ =	97.86 psi
$f_2 = f_{2V} + f_{2M}$ =	-48.93 psi
ϕ = (ACI 9.3.2.4) =	0.70
$f_{P\ allow} = 0.85 * \phi * f_c'$ =	1785.00 psi
$x = f_1 * L / (f_1 - f_2)$ =	30.00 in
$R_1 = 0.5 * f_1 * x * B$ =	32.29 kips

Compare Transverse condition to Longitudinal condition

	TRANS.	.LONG.		CONTROLS
M_U @ Top of Socket	253.14	0.00	ft-kips	TRANS.
V_U @ Top of Socket	25.49	24.22	kips	TRANS.
$R_1 = 0.5 * f_1 * x * B$ =	133.46	32.29	kips	TRANS.
$x = f_1 * L / (f_1 - f_2)$ =	23.69	30.00	in	LONG.
$f_1 = f_{1V} + f_{1M}$ =	512.11	97.86	psi	TRANS.

By Comparison :

Use "E" ties same as "F" ties.

Check Shear Transfer @ top of Footing

$V_U = 0.5 * V_U$ @ socket =	12.11 kips
ϕ = (ACI 9.3.2.3) =	0.85
μ = (ACI 11.7.4.3) =	1.00
$A_{vf} = V_U / (\phi * f_Y * \mu)$ =	0.24 in^2
A_{vf} PROV'D = (6 - #6) =	2.64 in^2

Use same reinforcement as Transverse direction except as noted above.

Leveling Plate Design

Ultimate weight of P.S. column =	41.19 kips	
Impact @ 50% of weight =	20.60 kips	
P_U = Ult wt. + Impact =	61.79 kips	
ϕ = (ACI 9.3.2.4) =	0.70	
A_1 = 6" square plate =	36 in^2	
$A_2 = B^2$ =	484.00 in^2	
$f_P = P_U / A_1$ =	1.72 ksi	
$F_{P\ allow} = 0.85\phi(f_c')\sqrt{\frac{A_2}{A_1}}$ =	6.54 ksi	OK